NIHR | National Institute for Health and Care Research

Public Health Research

Check for updates

Synopsis

The impact of vaping and smoking on nicotine intake and toxicant exposure among youth in England compared with youth in North America

Ann McNeill,[1*] Deborah Robson,[1] David Hammond,[2] Jessica L Reid,[2] Maciej L Goniewicz,[3] Ashleigh C Block,[3] Richard J O'Connor,[3] Maria Nikolaidou,[1] Eve Taylor,[1] Katherine East,[1] Eileen Brobbin,[1] Sarah Aleyan,[1] Marzena Orzol,[1] Kirstie Soar,[1] John Robins[1] and Leonie S Brose[1]

[1]Addictions Department, Institute of Psychiatry, Psychology and Neuroscience (IoPPN), King's College London, London, UK
[2]School of Public Health Sciences, University of Waterloo, Waterloo, ON, Canada
[3]Roswell Park Comprehensive Cancer Center, Buffalo, NY, USA

[*]**Corresponding author** ann.mcneill@kcl.ac.uk

Published April 2026
DOI: 10.3310/GJAM3822
Volume 14 • Issue 7

Abstract

Background: Youth vaping prevalence varies across countries and may be related to differing regulations/products. The emergence of cheap disposable vapes and high-concentration nicotine salts heightened concerns related to youth's ease of access, dependence and potential health risks.

Objectives: We examined youth in England versus Canada and the United States and:

1. how patterns of vaping/smoking varied, given the countries' different regulatory frameworks
2. nicotine and potential toxicant exposure in youth who vape, smoke or do neither
3. in youth who use salt and free-base nicotine
4. respiratory symptom reporting.

Design, methods, setting and participants: Objectives 1 and 4: Repeated cross-sectional surveys (International Tobacco Control Youth Tobacco and Vaping Surveys), England, Canada, United States, 2017–22, *N* = 1355–39,214, 16- to 19-year-olds, or 16- to 29-year-olds, depending on analysis.
Objectives 2 and 3: International Tobacco Control Biomarker Sub-Study, England, Canada, United States, 2019–22, 16- to 19-year-olds, *N* = 364 and England-only Biomarker Sub-Study, 2022–3, 16- to 19-year-olds, *N* = 201. Past-week users and past 30-day non-users were tested.

Interventions: None, comparisons based on vaping/smoking status.

Main outcome measures: Objective 1: Vape flavours, nicotine concentration, product types, brands used.
Objectives 2 and 3: Urinary biomarkers, normalised for creatinine; tobacco-specific nitrosamine NNK (NNAL); volatile organic compounds (VOCs): acrolein (3HPMA), acrylamide (2CaHEMA), acrylonitrile (2CyEMA), benzene (PhMA), toluene (BzMA), xylene (24MPhMA); nicotine: cotinine, trans-3'-hydroxycotinine (3-HC), total nicotine equivalents.
Objective 4: Self-reporting any of 5 past-week respiratory symptoms (e.g. cough and dyspnoea).

Results: Objective 1: Usual flavours were unchanged after 2020 United States pod-based vape flavour restrictions. Youth used exempt brands/products. Simultaneously, disposable vape use increased. In England, in 2022, 48% of 16- to 29-year-olds who vaped in past 30 days used Elf Bar brands, mainly for subjective responses (e.g. flavour/taste), rather than quitting smoking. Nicotine concentrations varied cross-country.

This synopsis should be referenced as follows:
McNeill A, Robson D, Hammond D, Reid JL, Goniewicz ML, Block AC, *et al*. The impact of vaping and smoking on nicotine intake and toxicant exposure among youth in England compared with youth in North America. *Public Health Res* 2026;**14**(7). https://doi.org/10.3310/GJAM3822

Objectives 1, 2 and 3: Compared to smoking tobacco (exclusive or alongside vaping), exclusive vaping was associated with: similar nicotine exposure (those using nicotine salts had higher levels of nicotine metabolites vs. free-base/unknown); lower exposure to NNK, acrolein, acrylamide and acrylonitrile, but higher toluene exposure (than dual use). Compared with not vaping/smoking, exclusive vaping was associated with similar exposure to acrolein and acrylonitrile and higher exposure to toluene and acrylamide (past 24-hour sensitivity analysis). Benzene and xylene biomarkers were detected in < 5% of urine samples. Some country-level biomarker differences were observed.
Objective 4: Vaping was associated with higher respiratory symptom reporting than not vaping/smoking. Youth who smoked and vaped had higher odds of symptoms than those only vaping. Using fruit, multiple or 'other' flavours was associated with higher odds of symptoms than tobacco flavours. Nicotine salt use was frequently unknown but may be associated with symptoms.
Limitations: Recall, misunderstandings and misreporting are possible. A subset of biomarkers was included, not all potential confounders were assessed and categorisation into vaping/smoking groups based on past-week behaviour does not fully account for past smoking exposure.
Conclusions: Pod flavour restrictions were ineffective. Youth were increasingly using disposable vapes containing nicotine salts. Those who vape were exposed to lower levels of toxicants than those who smoke, but a few toxicants were higher compared to youth who did not vape/smoke. Self-reported past-week respiratory symptoms were also higher in those who vaped than those not vaping/smoking and were related to flavours.
Future work: The rapidly evolving nicotine vape market needs ongoing survey/biomarker research.
Funding: This synopsis presents independent research funded by the National Institute for Health and Care Research (NIHR) Public Health Research programme as award number NIHR130292.
A plain language summary of this synopsis is available on the NIHR Journals Library Website https://doi.org/10.3310/GJAM3822.

Introduction

Rationale for research and background

Youth vaping (nicotine e-cigarette use) is a growing global concern. In 2019, when this study was designed, vaping among youth was sharply increasing in Canada and the USA,[1,2] but it remained stable in England despite rising experimentation.[1,3] The reasons for these differences across these high-income countries were unclear but could have related to varying regulatory frameworks that also influenced product types used.

At the time, all three countries prohibited nicotine vapes to under-18-year-olds, with some North American states/provinces having higher age limits. However, other regulations differed considerably. England adhered to the European Union Tobacco Products Directive, requiring product notification to the Medicines and Healthcare products Regulatory Agency, limiting nicotine concentration (20 mg/ml) and restricting some ingredients/flavours and marketing channels. In the USA, regulations were minimal, with no nicotine limits or enforced product notification,[4] though changes were planned for May 2020. Canada legalised nicotine vaping in 2018 with a 66 mg/ml nicotine limit,[5] following a poorly enforced ban.[6] Subsequent regulatory changes in the USA in 2020 included federal restrictions on the sale of flavours other than tobacco and menthol in pod/cartridge vapes (but not refillable tank vapes or disposable e-cigarettes), providing evaluation opportunities for this study.

Nicotine salt technology raised concerns about youth exposure due to its ability to deliver higher nicotine concentrations with reduced upper airway irritation,[7] enhancing the sensory experience. Popularised in 2018, particularly in North America,[1] vapes using nicotine salt e-liquids contained up to 59 mg/ml nicotine in countries without a 20 mg/ml limit.[8] Other health concerns included the 2019 US outbreak of e-cigarette or Vaping product use Associated Lung Injury, initially attributed to nicotine vapes, but later linked to vitamin E acetate-contaminated tetrahydrocannabinol products.

Urgency for youth vaping research increased amidst these concerns. Biomarker studies offered a way to assess health risks, but had been largely limited to US adults, with a few studies of US youth but no data on English youth. Cross-country comparisons of regulatory environments could help explain differing vaping rates.

This National Institute for Health and Care Research (NIHR) study extended the International Tobacco Control (ITC) Youth Tobacco and Vaping Surveys[9] and ITC Biomarker Sub-Study, led by Professor David Hammond. It examined the associations between regulations and vaping and

smoking patterns, nicotine/toxicant exposure and health effects, including respiratory symptoms.

Objectives

1. To investigate how patterns of vaping and smoking among youth in England compare with patterns in Canada and the USA, which have different e-cigarette regulations.
2. To measure levels of exposure to nicotine and potential toxicants and how these differ between *youth who vape*, *smoke or do neither* in England compared with the other countries.
3. To explore if exposure to nicotine and potential toxicants differs between *youth who vape nicotine salt e-liquids and youth who vape free-base nicotine* in England compared with the other countries.
4. To explore if vaping affects respiratory symptom reporting among youth across England, Canada and the USA.

Methods

Please see protocol July'23 v5.[10] Strobe v4 guidelines were used to guide the write up of the study.[11]

This study included several components; *Figure 1* presents a summary of the study components, the objectives addressed and the papers published relating to the objectives. We used repeat cross-sectional surveys in Canada, England and the USA with a nested biomarker substudy drawing from some surveys and a separate biomarker substudy in England only.

Details on sample size justification and statistical methods are given in the individual papers, but these are summarised briefly below.

We established a User Involvement Group of youth specifically for this study – see *Patient and public involvement* for details.

International Tobacco Control Youth Tobacco and Vaping Surveys (International Tobacco Control Youth Surveys)

The ITC Youth Surveys are online repeat cross-sectional surveys designed to examine the use of e-cigarettes and smoking among youth in three countries with different policy environments: the USA, Canada and England.[9] The ITC Youth Surveys focused on 16- to 19-year-olds, an age group in which regular smoking, and likely regular vaping, becomes consolidated.[12]

The ITC Youth Surveys were originally funded by the National Institutes of Health (NIH) in the USA for annual administration from 2017 to 2020. In light of the increases in youth e-cigarette use observed in Canada,[1,2] the surveys were enhanced with funding from Health Canada to include additional semi-annual surveys in 2020 and 2021, as well as 4-week follow-up surveys. In 2022, the NIH-funded study was extended through to 2025, with the sample extended to include both youth aged 16–19 years and young adults aged 20–29 years in each of the three countries. Technical reports are available.[9]

The sample is recruited from the Nielsen Consumer Insights Global Panel and their partner panels, in Canada, England and the USA. Nielsen panels are recruited using both probability and non-probability sampling methods in each country. Respondents are recruited either directly or through their parents. After screening participants for eligibility, all potential respondents were provided with information about the study and were asked for consent. Respondents received remuneration in accordance with their panel's usual incentive structure, which could either include monetary awards or points-based rewards and/or chances to win monthly prizes.

The surveys include questions on the use of e-cigarettes (all product types) and smoking (including manufactured and rolling tobacco), dependence, urges to use and frequency of smoking, vaping and dual use. Questionnaires are available for each survey in the relevant technical report.[9]

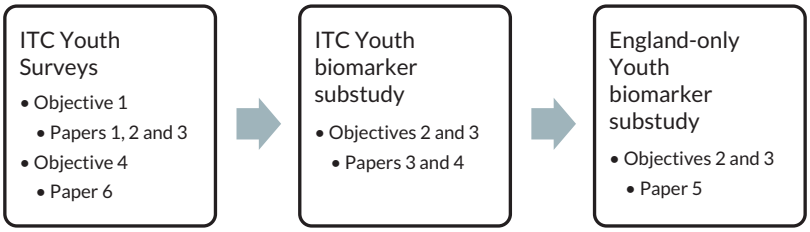

FIGURE 1 Data sources, objectives and peer-reviewed papers.

For survey outcomes, for example, a sample size of 4000 per country and using England's prevalence of vaping 20+ days per month (1.9%) from the 2018 survey gave us sufficient power to detect a difference of approximately 1.0% between countries within any single year (for a two-tailed test, where α = 0.05 and power = 0.8).

International Tobacco Control Youth Biomarker Substudy

Health Canada funding was also used to establish the biomarker substudy, which involved data collection from 2019 to 2022, involving participants initially surveyed in the ITC Youth Surveys [Wave 3 (September–October 2019), Wave 3.5 (February–March 2020), Wave 4 (August 2020), Wave 4.5 (February–March 2021) and Wave 5 (August–September 2021)].

Eligible respondents needed to pass the data quality question in the ITC Youth Survey, that is 'What is the current month?' (response options of all months, do not know, refuse; correct responses determined by current date ± 2 days); they also needed to belong to a panel that allowed this additional participation request, have parental permission to participate if under 18 years of age, be living at an address where they would be able to receive mail for at least the subsequent 4 weeks and provide their contact information, including postal address. They had to be in one of four groups:

1. people who vaped in the past week (but did not smoke)
2. people who smoked in the past week (but did not vape)
3. people who had vaped and smoked in the past week ('dual users')
4. non-users (no tobacco or vaping product or cannabis use in the last 30 days).

At the end of the ITC Youth Survey, eligible respondents who were interested in participating in the biomarker substudy received information about the substudy and were asked to provide their consent (and parent/legal guardian consent for those under the age of 18 years). Those who consented were sent a urine collection kit, instructions and shipping materials (including insulated container and cold packs) and a short questionnaire to collect data on past-week e-cigarette use, tobacco use and cannabis use, at the time of their urine collection. For youth who had vaped, measures also included: last e-cigarette brand (What brand was the *device* (please be as specific as possible); What brand was the *cartridge, pod or e-liquid*? (Please be as specific as possible); nicotine content and nicotine salt content (Did it contain nicotine? Yes/no/do not know); *If yes*, was it nicotine salt? (Yes/no/do not know). For biomarker samples taken from participants involved in Wave 3.5 onwards, they were also asked about flavour (What *flavour* was the cartridge, pod or e-liquid?); and those answering yes to containing nicotine were asked about nicotine concentration (What concentration of nicotine was it?) (Answer in mg/ml or %) with an option of do not know provided.

Participants were provided with instruction sheets on how to collect their urine sample, pack and return the sample and a short questionnaire. Ice packs and packaging material were provided and first-class mail or couriers ensured quick delivery. Materials were pre-labelled with numeric identification (ID) labels to ensure that unique participants could be identified while preserving anonymity during transit and storage. Kits from England were sent to King's College London using postage-paid envelopes, and once received, genetic material was removed from the urine samples by centrifuge within 7 days of receipt (removing genetic material is not required in Canada and the USA). Participants for whom samples were received were provided with an Amazon gift card. The samples were then stored in a −20 °C freezer until all the samples had been collected. They were sent to the Roswell Park Comprehensive Cancer Centre in Buffalo, New York, USA and were analysed to provide estimates of exposure to potentially harmful constituents across the four user groups in all three countries. Full details of the biomarkers measured, and analytical methods are provided in the papers and in *Table 1*. Roswell Park Comprehensive Cancer Centre was chosen to analyse the samples from all three countries to ensure comparability across countries.[13] Sample collection started in September 2019 and closed in January 2022.

Given the lack of available biomarker data for 16- to 19-year-olds, it was difficult to do power calculations for the biomarker substudies. We initially targeted a sample size of 180 participants for each user group across the three countries (n = 720) using power calculations for individual biomarkers. For example, for the biomarker for acrolein, 3-hydroxypropyl mercapturic acid (3-HPMA), 180 per user group was estimated to detect pairwise differences of 85 ng/g creatine [assuming a control mean of 327.8 ng/g, standard deviation (SD) 303.5 ng/g], for a two-tailed test where α = 0.05 and power = 0.8.

Components supported directly by National Institute for Health and Care Research funding

Respiratory symptoms

The NIHR funding enabled questions on the presence of respiratory symptoms to be added to the ITC Youth Tobacco and Vaping survey [in three surveys, Wave 4

TABLE 1 Biomarkers included in our studies

Testing group	Published papers	Parent compound	Classification/clinical relevance	Metabolite measured	Acronym (where appropriate)[14,15]	Unit	LLOQ (ng/ml)
Nicotine biomarkers	Hammond *et al.*[16] Taylor *et al.*[17]	Nicotine	Dependence,[18] reproductive or developmental toxicant.[19] Also considered as a cardiovascular toxicant[20]	Cotinine Trans-3'-hydroxycotinine TNEs (molar sum of cotinine and 3-HC concentrations)	3-HC TNE	ng/ml	5 (cotinine) 50 (3-HC) N/A (TNE)
TSNA	Hammond *et al.*[21] Taylor *et al.*[17]	4-(Methylnitrosamino)-1-(3-pyridyl)-1-Butanone (NNK)	Known human carcinogen (IARC group 1)[22]	4-(Methylnitrosamino)-1-(3-pyridyl)-1-butanol	NNAL	pg/ml	3.0
VOCs	Hammond *et al.*[21]	Acrolein	Probable carcinogen (IARC group 2A[22]); respiratory toxicant; cardiovascular toxicant (FDA HPHC[19])	3-Hydroxypropyl mercapturic acid	3-HPMA	ng/ml	30.0
		Acrylamide	Probable carcinogen (IARC group 2A[22])	2-Carbamoyl-2-Hydroxyethyl mercapturic acid	2CaHEMA	ng/ml	10.0
		Acrylonitrile	Possible carcinogen (IARC group 2B[22]); respiratory toxicant (FDA HPHC[19])	2-Cyanoethyl mercapturic acid	2CyEMA	ng/ml	5.0
		Benzene	Known carcinogen (IARC group 1[22]); cardiovascular toxicant; reproductive or developmental toxicant (FDA HPHC[19])	Phenyl mercapturic acid	PhMA	ng/ml	2.0
		Toluene	Respiratory toxicant; reproductive or developmental toxicant (FDA HPHC[19])	Benzyl mercapturic acid	BzMA	ng/ml	2.0
		Xylene	Not classified by IARC (group 3[22]); neurological effects[23]	2,4-Dimethylphenyl mercapturic acid	24MPhMA	ng/ml	2.0
Control	Hammond *et al.*[16] Hammond *et al.*[21] Taylor *et al.*[17]	Creatine	Indicator of urine dilution	Creatinine (ref range 1.2–346.5)		mg/dl	

FDA, US Food and Drug Administration; HPHC, harmful or potentially harmful constituent; IARC, International Agency for Research on Cancer; LLOQ, lowest limit of quantitation; N/A, not applicable; NNAL, 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanol; TNE, total nicotine equivalent; TSNA; tobacco-specific nitrosamine; VOC, volatile organic compound.

This synopsis should be referenced as follows:
McNeill A, Robson D, Hammond D, Reid JL, Goniewicz ML, Block AC, *et al.* The impact of vaping and smoking on nicotine intake and toxicant exposure among youth in England compared with youth in North America. *Public Health Res* 2026;**14**(7). https://doi.org/10.3310/GJAM3822

(August 2020), Wave 4.5 (February–March 2021) and Wave 5 (August–September 2021)]. These examined whether respondents had experienced various respiratory symptoms in the past week, including shortness of breath, wheezing, chest pain, a cough and the presence of phlegm.

England-only Youth Biomarker Sub-Study

The coronavirus disease (COVID) pandemic and consequent periodic lockdowns heavily disrupted the original biomarker substudy in England, as participants had to go to the post office to return their urine samples, and recruitment targets were not met (e.g. only 14 youth who vaped exclusively in the past 7 days provided urine samples). We therefore carried out an additional community biomarker substudy in England only in 2022 and 2023. We aimed to recruit an additional 200 16- to 19-year-olds (approximately 40 people who smoked, 40 people who vaped, 40 people who vaped and smoked, 40 non-users and an additional 40 people who vaped nicotine salt products). The aims, research questions, shipping, storage and analysis were the same as in the original study, but recruitment methods differed and were informed by our User Involvement Group, who also codeveloped the recruitment and study materials and helped to promote the study.

We recruited using a national social media campaign involving a static advertisement and a short video (approximately 30 seconds) produced by our User Involvement Group to promote the study. Social media targeted the specific age and interest groups on Twitter (Twitter, Inc., San Francisco, CA, USA) and King's College London Instagram account. These recruitment routes and methods were identified based on social media marketing statistics (e.g. study[24]) for the participant target age group. Additionally, recruitment involved placing advertisements, posters and flyers in Sixth Form Colleges, Colleges of Further Education and universities in London and the Northeast of England, advertisements in local vape shops and through the local press as well as via the King's College London Hype project website and social media accounts (the Hype project is an online recruitment and resource platform for young adults).[25] Advertisements included contact details and a quick response code for the research project web page. The study web page provided additional details, including a longer video clip (approximately 2 minutes), which was also produced by the User Involvement Group, explaining the study processes, and a participant information sheet and broad eligibility details.

Interested participants then (1) completed a consent form and screening questionnaire [via Qualtrics® (Qualtrics, Provo, UT, USA)] to check the eligibility criteria, provide contact information and receive more information and a unique participant ID code number; (2) completed a short video call to confirm eligibility and capacity to consent and provision of postal address; (3) completed an online questionnaire (via Qualtrics) assessing demographics, smoking and/or vaping behaviours, including flavours used, nicotine concentrations, brand and nicotine salt or non-salt e-liquid; (4) provided a urine sample at home and (5) completed an additional brief questionnaire at the time of their urine collection. Urine kits and the final brief questionnaire labelled with the participant's unique ID code were posted to eligible participants to complete at home and were then posted back to King's College London researchers using post-boxes (preferably priority post-boxes) to avoid having to go to post offices. The brief questionnaire was similar to that used in the ITC Youth Biomarker Sub-Study, including measures on brand, flavour, nicotine content, concentration and nicotine salts. In this substudy, however, few people who vaped knew whether they were vaping nicotine salts, and brand information was used to derive nicotine salt usage instead.

Eligibility criteria were as in the main study, but only personal consent was needed (youth over the age of 16 years are presumed to be capable of providing informed consent).[26] Participants of the community biomarker substudy received a £50 Love2Shop electronic voucher. Recruitment started in September 2022 and ended in December 2023.

Samples were sent initially to King's College London before sending to Roswell Park Comprehensive Cancer Center following identical procedures to the Biomarker Sub-Study above.

The power calculation for one-way analysis of variance tests demonstrated that $N = 40$ per group was sufficient to detect overall differences of an effect size of around 0.32 (Cohen's medium effect size) for 3-HPMA, where $\alpha = 0.05$ and power = 0.8. We targeted 200 participants in total [40 participants who smoke, 80 who vape (including 40 vaping nicotine salts), 40 participants who smoke and vape and 40 who do not vape or smoke].

Analysis

Table 2 includes information about data sources, design and participants. In brief, survey data were weighted and analysed using descriptive statistics and regression analysis; biomarker analysis included normalisation for creatinine and descriptive and regression analysis. Preregistration links for Papers 3–6 provide more details on analysis (see *Table 2*).

TABLE 2 Summary of papers

Paper	Data source and design	Research question	Project objective	Pre-registration and analysis summary	Participants	Principal findings
1. Hammond D *et al.*[27]	Repeat online cross-sectional ITC Youth Tobacco and Vaping Surveys, five waves from 2017 to 2020 (July–August 2017, August–September 2018, August–September 2019, February–March 2020, August 2020 survey waves included)	To examine the impact of US partial ban on flavours other than menthol and tobacco (in pod-based e-cigarettes, but not e-liquids, or disposable e-cigarettes) introduced in February 2020	Objective 1	No pre-registration Separate logistic regression models for each flavour/device type were used to test effects of time and country, adjusting for socio-demographics. Contrasts were used to group waves as pre-implementation, during implementation and post implementation	*N* = 9512 youth who vaped in the past 30 days, aged 16–19 years from England (*n* = 2262), Canada (*n* = 3331) and the USA (*n* = 3919)	Usual flavours used by youth who vaped in the USA were unchanged after the 2020 restrictions on pod-based e-cigarettes. Youth used brands and product types exempt from the restrictions There was a rise in the use of the new disposable e-cigarettes during the study
2. East K *et al.*[28]	Repeat online cross-sectional ITC Youth Tobacco and Vaping Surveys, wave 2022 only carried out in August/September 2022	To examine youth and young adults' use of Elf Bar (a highly popular brand of disposable vapes) sociodemographic characteristics and dependence indicators and reasons for use over other brands	Added in the light of Paper 1 (Objective 1) to research in more detail disposable use in England	No pre-registration Proportion of sample who reported currently using Elf Bar most often compared with other brands. Logistic regression models used to predict currently using Elf Bar examining sociodemographics and dependence indicators. Logistic regressions compared proportions for each reason	*N* = 1355 youth aged 16–29 years from England who vaped in the past 30 days, reported having a usual e-cigarette brand and had no missing data on one of the dependence measures	Elf Bar e-cigarettes were the most often used brand by approximately half of 16- to 29-year-olds who vaped in past 30 days in England in 2022 (48.4%). Elf Bar may have some of its own unique customer demographics and appeal. Using Elf Bar over other brands was associated with vaping in the afternoon rather than upon waking. Elf Bar was mainly chosen over other brands for subjective responses (e.g. flavour/taste) rather than for quitting smoking.
3. Hammond D *et al.*[16]	Cross-sectional analysis of ITC Youth Tobacco and Vaping Survey with at-home urine samples, September 2019–January 2022 (ITC Biomarker Sub-Study)	To examine differences in exposure to nicotine: by vaping and smoking status; across countries and by nicotine content and form	Objectives 1 and 2	Pre-registration available.[29] Urine samples tested for two nicotine metabolites (cotinine and trans-3'-hydroxycotinine), normalised for creatinine concentration. TNEs were calculated. Values below assay limit of quantitation (LOQ) imputed using formula LOQ/√2 with extreme values excluded. Logistic regression models conducted for each biomarker as reported in paper	*N* = 364 (17 in 2019, 257 in 2020, 89 in 2021, 1 in 2022). England (*n* = 131): 14 exclusively vaped in past week, 33 exclusively smoked past week, 27 vaped and smoked past week, 57 used neither product in past 30 days. Canada (*n* = 129): 35 exclusively vaped, 16 exclusively smoked, 26 vaped and smoked, 52 used neither product USA (*n* = 104): 24 exclusively vaped, 19 exclusively smoked, 24 vaped and smoked, 37 used neither product	Exclusively vaping in the last week was associated with similar nicotine metabolite levels compared to smoking tobacco (exclusive and when used alongside vapes). Among youth who exclusively vaped, those who reported using nicotine salts had higher levels of nicotine metabolites (vs. freebase products or where participants did not know). Among those who exclusively smoked, nicotine concentrations were lower in Canada than in England

continued

This synopsis should be referenced as follows: McNeill A, Robson D, Hammond D, Reid JL, Goniewicz ML, Block AC, *et al.* The impact of vaping and smoking on nicotine intake and toxicant exposure among youth in England compared with youth in North America. *Public Health Res* 2026;**14**(7). https://doi.org/10.3310/GJAM3822

DOI: 10.3310/GJAM3822

TABLE 2 Summary of papers (*continued*)

Paper	Data source and design	Research question	Project objective	Pre-registration and analysis summary	Participants	Principal findings
4. Hammond D *et al.*[21]	Cross-sectional analysis of ITC Youth Tobacco and Vaping Survey with at-home urine samples September 2019–January 2022 (ITC Biomarker Sub-Study)	To examine differences in toxicants (NNK and six VOCs) by vaping and smoking status; and across countries	Objectives 2 and 3	Pre-registration available.[29] Urine samples tested for metabolites of a TSNA (NNAL, a metabolite of NNK), and six VOCs [acrolein (3-HPMA), acrylamide (2CaHEMA), acrylonitrile (2CyEMA), benzene (PhMA), toluene (BzMA) and xylene (24MPhMA)]. Biomarkers normalised for creatinine. Values below assay limit of quantitation (LOQ) imputed using formula LOQ/√2 with extreme values excluded. Logistic regression models conducted for each biomarker as reported in paper	Same as in Paper 3	Youth who exclusively vaped had similar exposure as no vaping/smoking except for one VOC (toluene) and lower exposure than smoking or dual use except for toluene Higher NNAL levels were observed among US youth who smoke compared with youth in England and Canada
5. Taylor EV *et al.*[17]	Cross-sectional analysis of survey and urinary biomarker data. England-only Biomarker Sub-Study. September 2022–December 2023	To investigate the associations between levels of urinary nicotine metabolites and NNAL between youth aged 16–19 years in England by vaping and smoking status	Objective 3	Pre-registration available.[30] Biomarker testing as in Papers 2 and 3. Linear regression models as described in paper	*N* = 201 youth Exclusively vape (not smoked in past week), *N* = 83 Exclusively smoke cigarettes (not vaped in past week), *N* = 9 Smoke cigarettes and vape ('dual use') in past week, *N* = 55 Neither vape nor smoke cigarettes in the past week, *N* = 54	Youth who vape are exposed to similar levels of nicotine compared to those who smoke or who dual use. NNAL exposure among youth who vape is much lower than those who smoke, and indistinguishable from youth who do not vape or smoke Almost all youth who vaped in this study were vaping the new disposable products and nicotine salts
6. Brose LS *et al.*[31]	Cross-sectional analysis of survey data ITC Youth Online Survey pooled between 2020 and 2021. August 2020, February/March 2021, August/September 2021 waves included	Reporting any of five respiratory symptoms in the past week (shortness of breath, wheezing, chest pain, phlegm, cough) by vaping and smoking status and across countries	Objective 4	Pre-registration available[32] Proportions reporting any symptoms overall, by sample characteristics, and those not used other products. Bivariate logistic regression models assessed outcomes as described in paper	*N* = 39,214 youth aged 16–19 years from England, Canada and the USA	Vaping was associated with increased reporting of symptoms. Youth who both smoked and vaped had higher odds of experiencing symptoms compared to those who only vaped or only smoked. Users of fruit, multiple, or 'other' flavours reported higher odds of respiratory symptoms compared to those using tobacco flavours. Using nicotine salts was potentially associated with symptoms

2CaHEMA, 2-carbamoyl-2-hydroxyethyl mercapturic acid, acrylamide; 2CyEMA, 2-cyanoethyl mercapturic acid, acrylonitrile; 24MPhMA, 2,4-dimethylphenyl mercapturic acid, xylene; 3-HPMA, 3-hydroxypropyl mercapturic acid, acrolein; BZMA, benzyl mercapturic acid, toluene; PhMA, phenyl mercapturic acid is for benzene; NNAL, 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanol; TNE, total nicotine equivalent; TSNA, tobacco-specific nitrosamine; VOC, volatile organic compound.

Results summary

An overview of the papers and key findings is given in *Table 2*.

Discussion/interpretation

Some material is reproduced with permission from Brose *et al.*[31] This is an Open Access article distributed in accordance with the terms of the Creative Commons Attribution (CC BY 4.0) licence, which permits others to distribute, remix, adapt and build upon this work, for commercial use, provided the original work is properly cited. See: https://creativecommons.org/licenses/by/4.0/. The text below includes minor additions and formatting changes to the original text.

Principal findings and achievements per project outcome

Overall, when we started the study, we did not anticipate the rapid changes in regulations and products in the three countries, which created considerable complexity, in addition to the coronavirus disease discovered in 2019 (COVID-19) pandemic, which started shortly after our study began. Nevertheless, with some changes to the planned work, we were able to meet our objectives and add to the growing literature on youth vaping.

Objective 1. Patterns of vaping and smoking in England compared with Canada and United States, which have different regulations

Three papers solely addressed this objective 1. Paper 1[27] assessed the impact of a regulatory change in the USA, in 2020, which restricted flavours in pod (or cartridge)-based e-cigarettes but not in other nicotine vaping product types; this regulatory change was not implemented in England or Canada. There was no significant change in the usual flavours used by youth who had vaped in the past 30 days in the USA after the restrictions came in. In August 2020, the vast majority (79%) of 16- to 19-year-olds who vaped in the USA still reported using a flavour prohibited in pods/cartridges, similar to the proportion in Canada (86%) and England (80%). Youth who vaped in the USA appeared to have circumvented the flavour restrictions by using product types exempt from the restrictions, most notably new disposable products. The use of disposables increased significantly in the USA (the most popular brand was Puff Bar), with smaller increases in the other two countries.

No 16- to 19-year-olds in England reported Elf Bar use in 2017 through 2020. A report of the ITC Surveys[9] indicated that in England, in 2021, 1% of 16- to 19-year-olds who vaped reported using Elf Bar most often, and the 2022 data indicated a steep rise, which we studied in more depth in our Paper 2.[28] Participants in the 2022 survey were 16- to 29-year-olds who reported vaping in the past 30 days, reported having a usual brand of e-cigarette and did not have missing data on the key dependence measure (time to first vape). Among these participants, approximately half (48%) reported Elf Bar as the brand they currently used most often. Elf Bar usage varied significantly by age, from 41% of 16- to 17-year-olds, to 60% of 18- to 19-year-olds and 47% of 20- to 29-year-olds. Elf Bar usage was also higher among those who were female, identified as White race/ethnicity, were students, did not own a vape and typically vaped 5–8 hours after waking. Most who vaped but had never smoked used Elf Bar (64%), although this was not significantly different from those who vaped and also currently (45%), formerly (42%) or experimentally (49%) smoked. Popular reasons for choosing Elf Bar over other brands were better flavour/taste (48%), less expensive (29%), easier to get (26%), smoother to inhale (24%) and popularity (23%); better for quitting smoking (10%) was the least frequently selected reason for choosing Elf Bar over other brands. Moreover, 'stronger nicotine hit' was endorsed as a reason for use by only 12% of those who used Elf Bar most often. An additional assessment comparing use of any disposables versus other vaping products led to similar findings.

Results raised concerns about the appeal of Elf Bar vapes and other disposables and that the products were chosen by youth and young adults for their subjective responses (e.g. flavour/taste) rather than for stopping smoking; however, this may be explained by fewer current/former smokers using Elf Bar than other brands. Dependence findings suggested that Elf Bar use was more experimental, but at the time of this study (2022), Elf Bar products were still relatively new to the market and users might not have transitioned to heavier dependent use. Findings also indicated how product usage can be quickly affected by product changes on the market, in this case, over a 12-month period.

Paper 3[16] is discussed in more detail below, and the findings indicated that youth were largely using vapes with nicotine concentrations concordant with individual country regulations (at the time of our research, England had a nicotine cap of 20 mg/ml, whereas the other two countries did not). However, while there was some evidence of lower nicotine exposure in England, differences were not statistically significant.

Objective 2. Exposure to nicotine and potential toxicants by smoking and vaping and by country and Objective 3. Exposure to nicotine and potential toxicants by vaping nicotine salt versus free-base nicotine and by country

Papers 3–5[16,17,21] focused on Objectives 2 and 3, whereas Paper 3[16] also focused on Objective 1.

Paper 3[16] focused on nicotine biomarkers across the three countries among users of one or both products in the past week compared with those not using either product in the past 30 days (collected from September 2019 to January 2022). This analysis found no significant differences in total nicotine equivalents (TNEs) between youth who exclusively vaped (n = 73; geometric mean = 3.10 nmol/mg creatinine, SD = 16.69), exclusively smoked (n = 68; geometric mean = 3.78 nmol/mg creatinine, SD = 18.00; p = 0.82) or both vaped and smoked (n = 77; geometric mean = 6.07 nmol/mg creatinine, SD = 19.08; p = 0.20) in the past week adjusting for creatinine, age, sex, country and cannabis use; all groups had higher concentrations than non-users (n = 146; geometric mean = 0.19 nmol/mg creatinine, SD = 1.14; p < 0.001 for all contrasts). Similar patterns of exposure were observed for cotinine and trans-3'-hydroxycotinine.

Among youth that exclusively vaped, TNE concentrations were similar for those who reported using products containing > 20 mg/ml nicotine (n = 33; geometric mean = 4.35 nmol/mg creatinine, SD = 18.25) and containing ≤ 20 mg/ml nicotine (n = 28; geometric mean= 5.13 nmol/mg creatinine, SD = 15.64; p = 0.21). Reporting using vaping products containing nicotine salts (n = 23) was associated with higher concentration (geometric mean = 10.78 nmol/mg creatinine, SD = 18.03) than reporting products without nicotine salts (n = 29; geometric mean = 2.72 nmol/mg creatinine, SD = 15.42; p = 0.03) or do not know (n = 14; geometric mean = 1.55 nmol/mg creatinine, SD = 15.01; p = 0.009).

No statistically significant differences were found between the three countries within smoking/vaping groups, adjusted as above, except significantly lower trans-3'-hydroxycotinine and TNEs in Canada versus England among those exclusive smoking. The proportion of youth who self-reported using above 20 mg/ml nicotine concentrations varied across the three countries: England 7.1% (n = 1), Canada 37.1% (n = 13), USA 79.2% (n = 19), which is broadly consistent with nicotine cap of 20 mg/ml in England and a similar nicotine cap introduced in July 2021 in Canada. However, while there was some evidence of lower cotinine concentration among those exclusively vaping in England compared to Canada and the USA, the comparisons were not statistically significant.

Sensitivity analysis examined whether there was any difference among those who exclusively vaped (n = 73) by their smoking history (20 never smoked, 41 smoked < 100 cigarettes, 12 smoked ≥ 100 cigarettes). In this sample, smoking history was not associated with any nicotine biomarker concentrations.

Paper 4[21] used the same sample as in Paper 3 but tested for metabolites of a tobacco-specific nitrosamine (TSNA), which was 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanol (NNAL), a metabolite of the TSNA NNK, and biomarkers of six volatile organic compounds (VOCs): acrolein (3-HPMA), acrylamide (2-carbamoyl-2-hydroxyethyl mercapturic acid), acrylonitrile (2-cyanoethyl mercapturic acid), toluene (benzyl mercapturic acid), benzene (phenyl mercapturic acid) and xylene (2,4-dimethylphenyl mercapturic acid). For the latter two biomarkers, benzene and xylene, > 95% of samples had concentrations below the detection threshold and concentrations could not therefore be compared between groups. Youth who vaped were exposed to lower levels of the remaining biomarkers compared to smoking/dual use, except for toluene which was higher than dual use. TSNA biomarkers among those who vaped exclusively were indistinguishable from those who did not vape or smoke. For VOCs, biomarkers for toluene were higher among youth who exclusively vaped compared to youth who neither vaped nor smoked; for the others, there were no significant differences. In a sensitivity analysis using past 24-hour use rather than past-week use, findings were similar except acrylamide was higher among those exclusive vaping compared with those not smoking or vaping, and NNAL and acrylonitrile were higher among those exclusively smoking than among those smoking and vaping.

Country differences were examined for TSNAs (because there was an a priori hypothesis about TSNA given different types of tobacco are used across countries) among youth who smoked (dual or exclusive) as previous studies showed US blends to be associated with higher exposure among adults; we also found youth in the USA who smoked had higher levels of NNAL than those in England or Canada.

Paper 5[17] studied a separate sample of youth from England in 2022–3, which included larger numbers who exclusively vaped in the past 7 days. Among participants who had vaped, most used a disposable type (77% among those who exclusively vaped and 76% among those who vaped and smoked), nicotine strengths of 11–20 mg (78%, 76%),

nicotine salts (94%, 88%) and fruit flavoured e-liquids (76%, 58%). The data indicated the inter-relationship between smoking and vaping in this age group. Of those who exclusively vaped, the majority had also smoked at some point (73%). Similarly, among those who exclusively smoked, all had tried vaping. Among participants who neither vaped nor smoked, most had never smoked (70%), and equal proportions had never vaped (44%), or had vaped but not in the past 30 days (44%).

Youth who had vaped in the past 7 days were exposed to similar levels of nicotine (n = 83, geometric mean = 5.09 nmol/mg creatinine), but significantly lower levels of NNAL, than youth who smoked (n = 9, geometric mean = 1.74, p = 0.426) or both smoked and vaped in the past 7 days (n = 55, geometric mean = 5.60, p = 0.953). For youth who vaped in the past 7 days, NNAL levels (geometric mean = 1.87) were significantly lower than those who smoked (geometric mean = 4.87, p = 0.001) or both smoked and vaped (geometric mean = 3.67, p < 0.001) but not significantly different from youth who did not vape or smoke (geometric mean = 1.84, p = 0.887). Findings were not affected by recent cannabis use or second-hand smoke exposure. Among youth who vaped, nicotine exposure, but not NNAL, was associated with vaping recency. Levels of nicotine among participants who had exclusively vaped in our sample were substantially greater than among samples from youth in England in Paper 3, but similar to USA and Canadian samples. This may be due to differences in the product types and e-liquids that youth used. We were unable to compare nicotine types or other product characteristics in this study due to lack of variation, as most of the sample used disposable vapes, 20 mg/ml nicotine concentrations and nicotine salt e-liquids.

There were no significant differences in nicotine biomarkers between the sample of youth who vaped but had never smoked (n = 19) when compared with those who smoked (n = 9) or smoked and vaped (n = 55).

Overall, Papers 3, 4 and 5[16,17,21] indicated that youth who exclusively vaped had similar nicotine metabolite levels to those who smoked, but significantly lower exposure to harmful TSNAs and most VOCs, except toluene. Smoking history did not appear to affect these findings, although sample sizes were small (only 20 in Paper 4 exclusively vaped but had never smoked, and this was 19 in Paper 5). Youth using nicotine salt products showed higher nicotine metabolite levels. These findings suggest that while vaping delivers nicotine comparably to smoking, even for those who have never smoked, it reduces exposure to several other toxicants, albeit not completely.

Objective 4. Respiratory symptoms

Paper 6[31] addressed objective 4 and explored the association between vaping and smoking status and any past-week respiratory symptoms (shortness of breath, chest pain, phlegm, cough, wheezing) among youth aged 16–19 years across the three countries. Overall, more than a quarter reported experiencing symptoms in the past week. Youth who only vaped reported symptoms (40%) more often than those who neither vaped nor smoked [25%, odds ratio (OR) = 0.67, 95% confidence interval (CI): 0.61 to 0.72], with similar frequency as those who smoked only (39%, OR = 0.97, 95% CI: 0.85 to 1.10) and less often than those who smoked and vaped (49%, OR = 1.26, 95% CI: 1.12 to 1.42). The reporting of symptoms was associated with the frequency of vaping. Among youth who vaped in the past 30 days, some characteristics of vaping products were also associated with respiratory symptoms. Using nicotine salts was associated with increased symptom reporting overall but not when removing participants who had not previously heard of nicotine salts. While there was no evidence of differential associations by usual vape product type when using any symptoms as outcome, disposables were associated with three of five individual symptoms (shortness of breath, chest pain and phlegm). Respondents who reported usually using multiple flavours were consistently more likely to report symptoms (51%) than those using tobacco flavours (34%), while fruit flavours and 'other' flavours were only associated if not excluding use of other products. There were no clear differences in respiratory symptoms across the three countries among youth who vape.

Contribution to existing knowledge

Regulations

Our research added to a growing evidence base about the impact of regulations on vaping (and smoking) behaviours. In the USA, a partial restriction on flavours (applied to e-cigarette pod/cartridge product types but not others) had limited effect. The main impact of these restrictions was a market-wide shift to disposable products among youth, not subject to the flavour restrictions, a trend which was subsequently seen in England (at the time of this particular study, most youth were using refillable products) and many other countries. At the time of publication of our manuscript, there had been much debate about the restrictions, given concerns in the USA about youth vaping. Our research was novel as it examined the use of flavoured e-cigarettes over a 4-year period before and after the ban using a unique quasi-experimental design comparing the USA to England and Canada who had not implemented the restrictions.

Our research also addressed flavours in e-cigarettes, which is an ongoing concern in relation to youth use. At the time of this research, in England, fruit flavours were the most popular among youth who vaped, but it was also popular among adults.[33,34] This was thought to be due to their greater appeal for youth and greater satisfaction for adults.[35,36] Research on the importance of flavours has continued to grow,[37,38] as well as concerns about both flavours and youth vaping, which are currently the subject of powers in the Tobacco and Vapes Bill[39] going through the UK Parliament. Literature on potential unintended consequences of flavour bans on e-cigarettes has also grown,[38,40] demonstrating the need to consider unintended consequences, such as the impact on cigarette smoking.

Our results indicated that youth were largely using vapes with nicotine concentrations in line with individual country regulations. While there was some evidence of lower nicotine exposure in England, differences were not statistically significant.

Products

Ours was the first research study, to our knowledge, to specifically examine the use of Elf Bar vapes among youth and young adults, despite their popularity. However, consistent with qualitative research[41] on use of disposables, and concerns expressed in the literature,[42] and by the UK government,[43] better flavour/taste, low cost and ease of access were common reasons for choosing Elf Bar, as well as disposables in general. These findings may reflect disposables retailing at low prices and often in multibuy offers, which need curtailing. There was little clear evidence that nicotine dependence was higher among youth and young adults who primarily used Elf Bar. However, our research on biomarkers indicated that nicotine exposure among youth who vape in England appears to have increased during a time when the use of disposables, such as Elf Bar, increased rapidly.

Across Papers 3, 4, 5 and 6,[16,17,21,31] many respondents were unsure if their products contained nicotine salts. Thus, any self-reported data on nicotine type in this and other studies should be interpreted with caution.

Biomarkers

While research among adults who vaped showed that nicotine levels could be similar to adults who smoked in experienced users,[42] this had not been demonstrated among youth. Only a few studies on youth nicotine exposure from vaping had been carried out prior to our research, all from the USA,[44–47] often with small sample sizes.

Prior literature had suggested e-liquids containing nicotine salts, rather than freebase nicotine, would be associated with higher levels of nicotine delivery comparable to levels from smoking.[48,49] Biomarker studies had found differences in nicotine exposure among adults using different e-cigarette models, which were suggestive of higher exposure among those using tank products.[50–52] One study of youth found higher nicotine metabolite levels among youth who vaped 'pod' products, most likely due to these containing nicotine salt e-liquids, compared to non-pod products with freebase nicotine.[45] Our findings, however, are some of the first to investigate exposure by a range of vaping product characteristics, especially among disposable products. Other factors will also play a role in nicotine exposure, such as individual differences in nicotine metabolism and intensity and frequency of use.[42]

Our findings among youth, of significantly lower levels of non-nicotine toxicants among people who vape compared to smoke, are similar to those among adults, which we reviewed in the 2022 Nicotine Vaping in England report.[42]

One study from the USA by Chaffee *et al.* found no difference in NNAL between youth who used e-cigarettes and those who smoked tobacco or did not use either. However, the small sample size of youth using e-cigarettes (n = 12) provided limited power.[46] Our biomarker studies have clearly shown levels of NNAL, that were indistinguishable between youth who vaped and youth who neither vaped nor smoked, and much lower levels among youth who vaped compared with youth who smoked.

Another US study reported higher levels of metabolites for several VOCs (specifically acrylonitrile, acrolein, propylene oxide, acrylamide and crotonaldehyde) among youth who vaped in the last 24 hours versus non-using controls.[52] In our study, there were no differences for acrylonitrile, acrolein and acrylamide between youth vapers and non-using controls; propylene oxide and crotonaldehyde were not assessed in our studies. However, in a sensitivity analysis that categorised smoking/vaping status based on past 24-hour use, acrylamide levels were higher among youth who vaped in the past 24 hours compared to youth who neither vaped nor smoked. Higher levels of toluene were observed between youth who vaped and non-using controls and youth who smoked and vaped. Possible reasons for the differences may be methodological and participant characteristics. For example, the Chaffee *et al.* study was conducted with a younger sample in which daily smoking was almost absent (< 1% of the sample) and no participants vaped daily.

Respiratory symptoms

Previous studies, primarily using the US Population Assessment of Tobacco and Health study survey, found little evidence of associations between vaping and acute

respiratory symptoms in youth once smoking status had been taken into account;[53–57] whereas in this survey, associations persisted when adjusting for smoking and when excluding those who smoked or vaped other products. The present survey data were collected more recently (2020–1 compared with 2015–8), and both the vaping product market and patterns in vaping behaviour have changed during this time.[42,58] In addition, the present survey asked about past-week, compared with past-year, symptoms. However, the present findings are in line with a longitudinal study which used 2015–8 data and assessed associations between past-year symptoms and past-30-day vaping.[59]

It is difficult to draw conclusions based on flavours, as many youth report using a range of flavours. Moreover, many youth likely use flavour mixes, such as blueberry lemonade or cherry ice, which are difficult to categorise within the current measures.

Strengths and weaknesses of the study

The ITC Youth Surveys included a large sample across the three countries, and data were weighted to improve representativeness for all ITC Youth Survey data. The study was the first to assess biomarkers in youth who vaped in England. The ITC Biomarker Sub-Study included over 100 16- to 19-year-olds from each country, but sample sizes for the four user groups within each country were smaller, including only 14 who vaped exclusively in England. We mitigated this by carrying out a separate biomarker study in England in which we recruited 201 youth; this included 83 who exclusively vaped, although we were able to recruit only 9 youth who exclusively smoked (although the ITC Biomarker Sub-Study recruited 33 youth who smoked exclusively). Nevertheless, it would be good for our findings to be replicated with larger sample sizes, as the small sample sizes in our study limited the strength of comparison between the user groups.

There are general limitations associated with assessing biomarkers of exposure and their associations with smoking and/or vaping. All biomarkers are subject to potential confounders, such as the presence of some VOCs in household products.[60] Although urines were self-collected, and this could have led to protocol violations, we had used the underlying methodology previously with adults;[61] this method reduced participant burden and likely increased participation and the geographical reach of study and importantly, enabled the study to go ahead in times of COVID-19 restrictions. For the biomarker sample analyses, we categorised smoking and vaping status based on the past 7 days, which seemed to be an appropriate period for estimating recent exposure, although this does not fully account for smoking and vaping histories, which can be highly variable among youth. Approximately, two-thirds of the ITC Biomarker Sub-Study occurred after the onset of the COVID-19 pandemic, which may have affected patterns of use and hence biomarker levels. Our community study was carried out after the COVID-19 lockdowns, so it is unlikely to have been affected by this. Finally, we were able to assess only a subset of potential biomarkers involved in vaping and smoking, but this is consistent with most other biomarker studies to date.[42]

The questionnaires were self-completed, which can lead to mistakes and potential misunderstandings of the questions. However, we assessed key measures with our User Involvement Group. Additionally, some youth did not know whether their vape contained nicotine or not, nor whether they were using nicotine salts. Furthermore, some measures could be under-reported; for example, reported use of cannabis, while high in our sample, may have been under-reported due to legal concerns. We took reported cannabis use into account in most of our analyses, but any under-reporting would affect biomarkers and respiratory symptoms. Respiratory symptoms were only assessed in the past week; considering longer-term (e.g. past-year) symptoms and longitudinal research assessing symptoms over time would increase confidence in the findings.

Take-home messages

Our findings on the impact of pod-/cartridge-specific flavour restrictions and changes in types of e-cigarettes used during the study highlighted the versatility of the e-cigarette market; accordingly, flavour restrictions and other product standards are likely to have a greater impact if they are applied across all e-cigarette market segments. However, other studies have shown potential unintended consequences of flavour bans on youth smoking, which should be taken into consideration when considering such regulations. Furthermore, we did not consider the impact on adults, and there may be negative consequences of vape flavour restrictions for adults who smoke.

Youth who vaped had similar levels of nicotine to youth who smoked, similar to previous findings with adults who vape or smoke. Nicotine salt products may expose youth to higher nicotine levels, which could affect their potential for dependence.

Youth who vaped had lower levels of toxicants compared to youth who smoked, for TSNAs and most VOCs that were assessed in our studies.

Vaping appears to be associated with an increased reporting of respiratory symptoms, similar to smoking.

Challenges faced and limitations

Our study started in May 2020, just after the COVID-19 pandemic outbreak. At first, we were optimistic that our study would be unlikely to be affected, given surveys were online and urine samples were to be self-collected by the participants and posted back to the laboratories. Additionally, we investigated the risks of COVID-19 transmission through urine, and it was deemed to be very low/non-existent, but we included a red warning card to participants to not post their urine samples if they had COVID symptoms and/or were self-isolating/in quarantine. However, we did not anticipate how long the COVID-19 pandemic would last and the multiple lockdowns that took place in England (and Canada). As some urine samples had to be posted by visiting post offices, we were unable to send sample kits to participants during lockdown periods to discourage them from breaking lockdown rules. This was quite complicated and time-consuming, given lockdowns varied across the country. Overall, data collection took additional time and although we managed to collect 131 samples from England, we were able to collect only samples from 14 youth who exclusively vaped. Given the uniqueness of the biomarker data from youth in England, we secured permission to extend the Biomarker Sub-Study in England using separate recruitment methods, but with the same survey questions, biomarker sampling and measurement procedures. Again, this was time-consuming to run and we experienced multiple delays in the ethics review process.

Once in the field, our recruitment processes worked smoothly, but on this occasion, we had difficulty recruiting youth who exclusively smoked despite extensive efforts on social media to target people who smoked. This might reflect the changing nature of nicotine use in this age group, where products can be used interchangeably.

At the time of the application and earlier stages of the project, our partners at Roswell Park Comprehensive Cancer Centre were anticipating that they would be able to measure selected polycyclic aromatic hydrocarbons (PAHs) in addition to VOCs and nicotine metabolites; however, by the time the samples were received, assessment of PAHs and VOCs from the England-only Youth Biomarker Sub-Study was not possible at that time due to ongoing equipment issues. We discussed this with our Study Steering Committee, and Tim Marczylo from the UK Health Security Agency (UKHSA) indicated that they might be able to support other biomarker analyses if we could secure the relevant ethical approvals and permission. We agreed that analysis of metals would be valuable, given recent studies indicating metals in vaping aerosols.[62] Roswell Park returned the unanalysed urine sample vials to UKHSA, and we aim to publish the findings subsequently.

During the course of the study, we also experienced multiple staff changes. COVID-19 caused the co-principal investigator Dr Sara Hitchman to leave England. Additionally, given the prolonged duration of the study, we had multiple staffing changes as research assistants sought longer-term positions or promotions. Across the lifetime of the project, a total of six researchers were employed on the one post (some casually), and there were some time periods without a full-time employed researcher.

Cannabis use was common in our samples and may have increased biomarkers of exposure and symptom reporting. For example, in Paper 5, ever cannabis use was common among participants who vaped (54.8%), smoked (88.9%) or dual used (88.9%), but it was less so for those who did not vape or smoke (22.2%). In Paper 6, 12.2% of respondents had smoked or vaped cannabis in the past 30 days. Cannabis vaping and/or smoking was controlled for in the models for Papers 3 and 4, strengthening the conclusions.

Finally, youth vaping was raising increasing concerns both in England and globally during our study. This impacted the regulatory frameworks in Canada, USA and England, likely leading to changes in product use over the study period.

Engagement with partners and stakeholders

We worked closely and strengthened relationships with our partners, particularly those at the University of Waterloo in Canada and Roswell Park Comprehensive Cancer Centre in the USA as well as the Social and Genetic and Developmental Psychiatry (SGDP) Centre at King's College London. As referred to above, there were many challenges during the study (lockdowns, lengthening fieldwork periods, staffing and changes in potential biomarkers being analysed), but we worked through these collectively.

We were also regularly in touch with policy colleagues to ensure our research would still be timely despite the delays. We are also leading a new 5-years NIHR Policy Research Unit in Addictions, which will ensure that we will be able to continue disseminating this research to relevant policy colleagues in Department of Health and Social Care (DHSC) over the next few years.

Through the participation of Dr Tim Marczylo in the Study Steering Committee, we have also further developed relationships with the UKHSA.

We are grateful to our Study Steering Committee who provided very helpful advice and support throughout our study.

Individual training and capacity-strengthening activities

While the involvement of several, sometimes short-term, researchers was challenging, it also meant that several early career researchers developed their skills in planning research, participant recruitment, additional analysis and dissemination. Close working with the Roswell Park Comprehensive Cancer Centre enabled all the UK researchers involved to strengthen their knowledge and interpretation of nicotine and vaping biomarkers, capacity for which in the UK is severely lacking.

Institutional capacity-strengthening

There is a severe lack of tobacco- and nicotine-related biomarker research conducted in the UK. Not only is this some of the first biomarker research done by our department, but it is also some of the first youth biomarker research done in the UK. This project has enabled our department to develop methods that are necessary to the future of tobacco and nicotine research. Methods developed can also be used outside of nicotine and tobacco research for multiple areas of population level toxicological monitoring. We have established interdepartmental working relationships with the SGDP which will support future research.

Moreover, the collaboration with the University of Waterloo, for which we are extremely grateful, expanded our wider team's (the Nicotine Research Group) capacity to work on the ITC Youth Surveys (and the wider ITC research), resulting in over 25 published research papers using ITC data, where a team member is a lead or coauthor (with many more in the pipeline).

Patient and public involvement

We established a bespoke User Involvement Group who were involved in the application stage and worked with us until data collection for the community substudy ended. Members included eight youth who were aged 18–21 years at the start of the study (seven males and one female). Two people had never smoked, four had tried smoking and vaping but had never engaged in either behaviour regularly and two smoked and vaped. Two of the group were also regular cannabis users. Five were in higher education and in part-time paid employment and three were in full-time employment. None of the group had previously been a participant in any kind of research study.

Application stage

Surveys

We provided the group with study materials from the ITC Youth Surveys, including: the surveys, participant information leaflet and the lay summary from the outline stage application form. We invited them to provide feedback and advice about the study concept, the acceptability and comprehensiveness of the wording and language used in the research materials and the appeal of the study. Most group members found the survey's purpose unclear and suggested more context at the start of the survey, especially about the researchers' goals. They felt the lay summary was clear and recommended this was included earlier in the survey. From the lay summary, they understood the importance of the research and believed it was of relevance to them and their peers. They found the survey questions easy to follow, though some participants suggested changing terms like 'remuneration' and 'regulation/regulatory bodies', which were amended by the research team. The original survey included an 'auction task'; most felt potential participants might feel deceived when they realised it was not real, possibly leading to them abandoning the survey. This was subsequently removed from the survey for this study. Members also expressed discomfort with the question about grades, especially when placed after questions on smoking, vaping and cannabis use. They felt it might make respondents feel 'judged', although said they would not mind if it appeared earlier in the survey. Half of the group members said they would take part in the survey if invited.

Urine collections and biomarker substudy materials

Members reviewed the information leaflet and urine kit instruction form. They made recommendations for the wording on the benefits of taking part in the study and instructions for where they had to post urine to and who to contact if they wanted to ask questions. These recommendations were actioned by the research team. All eight members said they would take part in biomarker study (motivated by the financial incentive for this part of the study).

Additional recruitment

As it became clear that we were struggling to meet our recruitment targets for the initial biomarker substudy, we consulted the involvement group members about additional/alternative recruitment strategies. They recommended increasing incentives from £30 to £50 (which was adopted) and in-person urine collection (which was considered but proved unfeasible).

England-only Youth Biomarker Sub-Study

When we progressed to the England-only Biomarker Sub-Study, the User Involvement Group members were instrumental in assisting with publicity and recruitment. They co-designed and produced (1) a short video clip for social media to advertise the study; (2) a longer, 2-minute video with more details about the study (this was embedded in the study website on a King's College London web page[63] and (3) an instructional video about consent, urine collection and postage. They were also involved in ensuring the paper versions of the instructions about urine collection and postage were youth-friendly. This was impactful and greatly assisted in recruitment. To support dissemination to participants, they are currently producing a brief video summarising the main findings which will then be shared with all those invited to the study, put on the study website and shared on our social media channels.

Equality, diversity and inclusion

Online surveys of youth in Great Britain[42] have shown little difference and inconsistent patterns between smoking and socioeconomic status, which is in contrast to much higher levels of smoking in more disadvantaged adults than those from more advantaged social groups. The national Smoking, Drinking and Drug Use (SDD) surveys in England[64] (11- to 15-year-olds) assess patterns by ethnicity; in 2023, smoking prevalence was 3% overall, 3% in White and Mixed ethnicity groups, but it was lower (1%) in each of the Asian and Black ethnicity groups; ethnicity data for vaping were not available. Given the lack of distinct patterns and the changing patterns of smoking and vaping over time, we endeavoured to recruit diverse samples to our studies.

The ITC Youth Surveys (Papers 1 and 2) recruited a diverse sample in terms of gender and race/ethnicity across the countries and weighting based on age, sex, geographic region and (in the USA only) race/ethnicity further enhance the representativeness of the sample, ensuring generalisability of findings.

The ITC Biomarker Sub-Study recruited a sample with a balance of young male (44%) and female (56%) participants. In relation to family's financial situation, 8% reported that they were 'not meeting basic expenses', with 32% 'living comfortably'. The majority of the sample (73%) were from White ethnic backgrounds, while 27% were from other/Mixed. Similar demographic patterns were observed in the England-only Biomarker Sub-Study sample; for example, 86% were from White ethnicity. There was a good representation across the regions in England.

In the England-only Youth Biomarker Sub-Study, there were no requirements on gender, ethnicity or other socio-demographic or economic grouping. However, we ensured a geographical spread across England to avoid adding more London-centric evidence and represented a diversity in relation to gender (59% female), ethnicity (73% White) and perceived family financial situation (20% 'not or just meeting basic expenses' and 43% 'living comfortably'). We also endeavoured to ensure that the products used by our sample reflect those shown to be popularly used in the ITC Youth Tobacco and Vaping Survey. As such, we believe our findings provide a good representation and are generalisable beyond our specific sample.

In terms of language used, we have adopted person-first language in our work, against occasional pushback, and our User Involvement Group helped us to develop language appropriate for the age group of our participants.

Our research team at King's College London has been all-female, supporting women in science and reflecting the team's background in psychology, nursing and adjacent fields (overwhelmingly female student populations). The team lacked diversity in ethnicity but included several European and American nationalities.

Impact and learning

Impact

Our findings are relevant to forthcoming legislation in the UK, but it may also influence regulations on vaping in other countries. The Tobacco and Vapes Bill[39] is currently going through the legislative process in the UK Parliament. This includes powers to regulate contents, flavours, packaging and product requirements for vapes; these powers will need to be further defined in subsequent regulations. The health impact of vaping among youth will be central to discussions defining the forthcoming regulation. Our findings on nicotine exposure, biomarkers of potential harm and self-reported respiratory symptoms will provide unique evidence which can help ensure evidence-based policy. We will submit evidence to any consultations related to vaping and smoking where relevant but are regularly in touch with officials in the DHSC about this research. We provided written and oral evidence to government consultations in 2023 and 2024, which shaped the previous version of the bill.

Additionally, Eve Taylor discussed this research in a documentary for Sky Kids on youth vaping. Ann McNeill in her regular discussions with the broadcast media has also alerted them that this biomarker research is ongoing (including the BBC), and they have expressed interest in the findings. Once the study is complete, we will share the summary and anticipate some media involvement. We further advertised the study on a recent Springer Medicine Webinar on youth vaping, which had 500 participants from across Europe.

Lessons learnt for future research

Recruitment

Recruiting participants for biomarker collection through existing market research panel surveys proved to be more difficult in England compared with the community study that we set up separately. Future research should consider establishing separate recruitment pipelines from the outset to save time.

Collaborator agreements

Setting up contracts with institutions was highly time-consuming. Exploring ways to streamline this process, potentially with support from funders, could be beneficial.

Age verification

Video call-based age verification was effective and well-received. Flexible platform options are essential to accommodate changing preferences across groups and time.

Participant information

Professionally made, video-based participant information was likely helpful and should be considered in future studies. We will produce a final video with our User Involvement Group on the findings when all are published.

Patient and public involvement importance

Patient and public involvement was crucial to study design and recruitment materials. For younger target groups, maintaining and refreshing patient and public involvement panels to match the target population is vital.

Sample collection

Postal urine sample collection was highly successful with minimal loss, providing a replicable method.

Shipping challenges

While international sample shipping has complexities, it is manageable with proper planning.

Related work arising from this study/ collaborations/further funding/future work

As discussed in *Institutional capacity-strengthening*, we have expanded existing and built new collaborations within and across institutions.

We are also leading a new 5-year NIHR Policy Research Unit in Addictions. While not currently running biomarker studies, this may become necessary to address future policy needs. We have also further developed relationships with the UKHSA through the participation of Dr Tim Marczylo in the Study Steering Committee. For future biomarker studies in England, where comparisons with biomarkers collected in other countries are not needed, the UKHSA may be able to provide biomarker analysis instead of shipping the samples to the USA. Indeed, since this work began, new research on biomarkers among people with a mental health condition who vaped and/or smoked or used neither product was initiated through the NIHR Health Protection Research Unit led by Imperial, in which we worked with the UKHSA.

The biomarker work has been taken into account in a new project entitled 'Scoping of Policy Impacts for Regulating E-cigarettes (SPIRE): a data and decision-analytic model mapping project' led by colleagues at University College London and the University of Sheffield (NIHR16683). This 9-month project, in which two coauthors from King's are involved, aims to identify data sources that can inform modelling to improve the co-ordination of future research efforts regarding vaping.

Dissemination for implementation

- Peer-reviewed papers as listed in *Award publications*.
- Evidence for government consultations as described above.
- Share information at UK E-cigarette research forum, quarterly meeting attended by UK and international policy-makers and practitioners.
- Video summary of findings for social media.
- Society for Research on Nicotine and Tobacco (SRNT) – Europe, 2023 meeting oral presentation: Associations between nicotine vaping and respiratory symptoms in young people in Canada, England and the USA.
- SRNT, 2023 annual meeting, San Antonio, poster presentation: Biomarkers of exposure among youth smokers, vapers, and dual users: associations with product characteristics and differences across countries.
- SRNT, 2024 annual meeting, Edinburgh, oral presentation: Demographics, dependence indicators,

and reasons for use of 'Elf Bar' among youth and young adults who vape in England.
- SRNT annual meeting 2025. Nicotine exposure among youth in England who vaped and/or smoked, including by vaping product characteristics.
- Paper 6 has been selected for an NIHR Alert for NIHR evidence which is currently going through revisions.

Implications for decision-makers

Paper 1 indicated little impact on the flavour choice of youth who vaped as a result of the partial ban on flavours in the USA in 2020, which was restricted to pod-/cartridge-like vaping products. Accordingly, any regulations need to apply across all product types to maximise the impact. However, care needs to be taken to consider potential unintended consequences of vaping flavour regulations, including wider research, which indicate potential increases in cigarette smoking among youth and adults.[65–67]

Paper 2 found that better flavour/taste, low cost and ease of access were common reasons for choosing Elf Bar vapes, as well as disposables in general, consistent with qualitative research.[41] This may reflect disposables retailing at low prices and often in multibuy offers. The findings supported concerns raised by government,[43] and disposable products will be banned from June 2025. However, the e-cigarette market is versatile, and manufacturers are already adapting towards disposable analogues that have many of the attributes of disposables that make them appealing but are rechargeable and refillable.[68] Accordingly, restrictions and other product standards are likely to have greater impact on reducing youth vaping if they are applied across all market segments; however, consideration of the impact on adults who smoke is required in case they are deterred from using vapes to stop smoking.

Papers 3 and 5 showed that vapes are efficient nicotine delivery products, and 16- to 19-year-olds who vape have similar nicotine intake as youth who smoke. Additionally, the nicotine salt e-liquids, now the most widely used, seem to enable more efficient nicotine delivery. This therefore, to some extent, over-rides the presence of nicotine caps and although there was some evidence nicotine exposure was lower in youth who vaped in England, the findings were not statistically significant. More efficient delivery of nicotine, however, increases the risk of dependence; importantly, regulation needs to focus not only on nicotine content in the liquid but also on the delivery to the user. Given that not all youth knew whether they were using nicotine salt forms of vapes, labelling on vape packaging could be improved to give more salience to the nicotine form used.

Paper 4 indicated that youth who vaped were exposed to much lower levels of the assessed toxicants than youth who smoked and were indistinguishable from youth who neither vaped nor smoked for some but not all toxicants. This indicates that regulation needs to be mindful of the differential in risk of different ways of consuming nicotine. Regulation of vapes must never increase smoking in youth.

Paper 6 found that vaping was associated with respiratory symptoms, independent of smoking. Using disposables, multiple flavours, and potentially, nicotine salts may be associated with reporting symptoms. This supports regulation to restrict attractiveness and accessibility of vapes to youth who do not smoke.

In summary, this research provides unique research from England on biomarkers associated with youth vaping compared with youth smoking and youth who neither smoke nor vape. We were also able to examine data comparing youth who vaped with youth who have *never* smoked, in England for this first time, but caution is needed because of small sample sizes in some analyses. Overall, this research suggests that nicotine exposure in youth who vape is similar to those who smoke, whether or not they previously smoked, but this research requires replication. The harms from smoking are caused by other constituents in tobacco smoke and not nicotine; however, issues around dependency on vaping and respiratory symptoms need greater exploration going forward.

The biomarker findings of toxicants, while limited to a subset of biomarkers, do indicate lower risks from vaping compared with smoking. Nevertheless, clear messaging to youth who do not smoke about not vaping should also be communicated.

Research recommendations

The most important recommendation is that research monitoring nicotine usage in youth should be ongoing. This is important, given the evolution of the nicotine vape market, as products are changing rapidly, raising concerns around health effects and addiction. Monitoring of youth smoking and vaping is done largely through funding from the non-governmental sector (e.g. Cancer Research UK), other than the SDD among Young People in England Surveys, which are conducted every 2 years and do not

capture the level of detail needed on products.[64] Our difficulties in recruiting youth who exclusively smoke illustrate how nicotine use might be changing in this age group such that products may be used interchangeably. It is, therefore, important that surveys can identify such changes by asking detailed questions on all nicotine products and frequency of use. Additionally, given the recent disposable vape ban and the forthcoming Tobacco and Vapes Bill[39] in the UK, assessing the impact of regulatory changes is needed through differently designed studies such as time series analyses or pre–post evaluations.

Research on youth vaping and/or smoking should include clinical tests to provide objective measures, for example of respiratory diseases.

Repeated monitoring of biomarkers for nicotine and exposure among youth is needed to capture how market and product changes affect exposure levels and potential health effects among youth. The nicotine exposure results across countries with different regulations were inconclusive so could benefit from further research.

Additionally, a greater range of biomarkers are needed to cover not just biomarkers specific to smoking but also those specific to vaping, as well as biomarker research on a wider range of nicotine-containing products, such as nicotine pouches.[42]

More research is still needed comparing youth who have never smoked to youth who have smoked previously, as this can affect biomarkers. More generally, research needs to be more comprehensive in assessing use of other potentially addictive products. Too often, research on nicotine, cannabis, alcohol or other drugs is conducted in isolation despite overlapping in the population.

As experimentation with nicotine use can start before the age of 16 years, inclusion of younger age groups would strengthen the available evidence.

Conclusions

Concerns about youth vaping are driving many regulatory changes but sometimes predicated on misunderstandings about the relative risks of smoking and nicotine vaping. Overall, vaping is viewed as less risky than smoking, but most of the research to date has been carried out on adults, rather than youth, and in the USA. Our biomarker research across three countries (England, Canada and USA) indicated that youth who vape are exposed to similar levels of nicotine but to lower levels of most of the toxicants that were studied compared to youth who smoke; in most cases, levels among those who vaped were similar to youth who did not smoke or vape. Levels of nicotine biomarkers were higher among youth who vaped nicotine salt products. Respiratory symptoms were more commonly reported among youth who vaped and/or smoked than those who did not use either product. Hence, we conclude that youth who smoke would be better off switching to vaping products to stop smoking completely. Youth who do not smoke or vape should not take up either product.

This and other research indicates that, when regulations are considered to control youth vaping, they must not drive youth to smoke nor prevent adults who smoke from switching to vaping. We also found that partial regulations, such as the flavour ban introduced only for one type of e-cigarettes (pods) in the USA, were not effective and were also associated with a switch to disposable products. In England, the use of one disposable product, Elf Bar, increased rapidly and seemed likely due to the greater appeal and accessibility of Elf Bar and other disposable products. While our findings pointed to greater nicotine exposure in users of nicotine salt-based products, users of Elf Bar who we studied appeared to have a lower dependence. The rapidly changing nature of the products used by youth during our study underscore the need for regular monitoring and research in this area.

Additional information

CRediT contribution statement

Ann McNeill (https://orcid.org/0000-0002-6223-4000): Conceptualisation, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualisation, Writing – original draft, Writing – reviewing and editing.

Deborah Robson (https://orcid.org/0000-0002-6447-3523): Conceptualisation, Funding acquisition, Methodology, Resources, Supervision, Validation, Visualisation, Writing – original draft, Writing – reviewing and editing.

David Hammond (https://orcid.org/0000-0001-8197-6010): Conceptualisation, Data curation, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – reviewing and editing.

Jessica L Reid (https://orcid.org/0000-0001-9909-6265): Data curation, Formal analysis, Methodology, Resources, Validation, Visualisation, Writing – reviewing and editing.

Maciej L Goniewicz (https://orcid.org/0000-0001-6748-3068): Conceptualisation, Data curation, Formal analysis, Methodology,

Resources, Validation, Visualisation, Writing – reviewing and editing.

Ashleigh C Block (https://orcid.org/0000-0001-9849-4421): Data curation, Formal analysis, Methodology, Resources, Validation, Visualisation, Writing – reviewing and editing.

Richard J O'Connor (https://orcid.org/0000-0003-0644-182X): Conceptualisation, Methodology, Resources, Validation, Writing – reviewing and editing.

Maria Nikolaidou (https://orcid.org/0000-0001-8106-7351): Investigation, Methodology, Resources, Validation, Writing – reviewing and editing.

Eve Taylor (https://orcid.org/0000-0003-3987-4760): Data curation, Formal analysis, Methodology, Resources, Validation, Visualisation, Writing – reviewing and editing.

Katherine East (https://orcid.org/0000-0001-9083-2131): Data curation, Formal analysis, Methodology, Resources, Validation, Writing – reviewing and editing.

Eileen Brobbin (https://orcid.org/0000-0001-8464-4847): Investigation, Methodology, Resources, Validation. Writing – reviewing and editing.

Sarah Aleyan (https://orcid.org/0000-0002-0842-1147): Investigation, Methodology, Resources, Validation, Writing – reviewing and editing.

Marzena Orzol (https://orcid.org/0000-0001-6624-5041): Conceptualisation, Methodology, Resources, Validation, Writing – reviewing and editing.

Kirstie Soar (https://orcid.org/0000-0001-5557-0922): Conceptualisation, Methodology, Resources, Validation, Writing – reviewing and editing.

John Robins (https://orcid.org/0000-0001-6182-4263): Methodology, Validation, Writing – reviewing and editing.

Leonie S Brose (https://orcid.org/0000-0001-6503-6854): Conceptualisation, Funding acquisition, Methodology, Resources, Supervision, Validation, Visualisation, Writing – original draft, Writing – reviewing and editing.

Anastasia Baffour-Kyei: Resources.

Esteban Tato-Barcia: Resources.

Olivia Cottam: Methodology, Resources.

Rhian Jones: Project administration.

Sara Hitchman: Conceptualisation, Funding acquisition.

Tim Marczylo: Methodology, Resources.

Acknowledgements

We thank Dr Sara Hitchman for helping to obtain funding and contributing to the conceptualisation of this project, when she was employed at King's College London. We also thank Dr Pete Driezen for his support for the ITC Surveys and with our original application, and Noel Leigh for his support with sample shipping and work at the Roswell Park Comprehensive Cancer Center. We are also grateful to Dr Tim Marczylo and Dr Oliva Cottam at the UK Health Security Agency for agreeing to analyse samples for metals which will be published subsequent to this synopsis. We are grateful to colleagues at the Social, Genetic and Developmental Psychiatry Centre at King's College London (Esteban Tato-Barcia and Anastasia Baffour-Kyei) for receiving urine samples and removing genetic material before shipment to the USA. Finally, we thank Rhian Jones for administrative support for this project.

Data-sharing statement

All data requests should be submitted to the corresponding author for consideration. Access to anonymised data may be granted following review.

Ethics statement

The ITC Surveys and Biomarker Sub-Study were reviewed and received ethics clearance through a University of Waterloo Research Ethics Committee (ORE#21847/31017, there are two numbers because the ethics was migrated to new system since study started and given this new number; initial approval 27 April 2017, but renewed annually and approved with every amendment, most recent approval date, 2 June 2025), King's College London Psychiatry, Nursing and Midwifery Research Ethics Subcommittee (#13748, 28 June 2019), and a Roswell Park Comprehensive Cancer Center Ethics Committee (STUDY00001085/P-522019, 24 October 2019).

Ethics clearance for the Community Biomarker Sub-Study and amendment were obtained from King's College London Psychiatry, Nursing and Midwifery Research Ethics Subcommittee (#29811, 1 September 2022; 13 July 2023).

Information governance statement

King's College London is committed to handling all personal information in line with the UK Data Protection Act (2018) and the General Data Protection Regulation (EU GDPR) 2016/679. Under the Data Protection legislation, King's College London is the Data Controller, and you can find out more about how we

handle personal data, including how to exercise your individual rights using the contact details for our Data Protection Officer here: www.kcl.ac.uk/professional-services/business-assurance/data-protection-introduction-data-losses-and-reporting.

Disclosure of interests

Full disclosure of interests: Completed ICMJE forms for all authors, including all related interests, are available in the toolkit on the NIHR Journals Library report publication page at https://doi.org/10.3310/GJAM3822.

Primary conflicts of interest: Ann McNeill, Deborah Robson, Maria Nikolaidou, Eve Taylor, Eileen Brobbin, Sarah Aleyan, Marzena Orzol, Kirstie Soar and Leonie S Brose report receiving grant support from the NIHR Public Health Research Programme NIHR 130292.

David Hammond reports receiving grant support from Health Canada's Substance Use and Addictions Program for this research.

Ann McNeill, David Hammond, Maciej L Goniewicz, Richard J O'Connor and Katherine East report receiving grant support from the US National Institutes of Health PO1 CA200512 during the current study.

Ann McNeill, Deborah Robson and Leonie S Brose report receiving grant support from the NIHR PRU in Addictions 2024–8 (NIHR 206123). Ann McNeill, Deborah Robson and John Robins were funded during this study by the NIHR Applied Research Collaboration South London (NIHR ARC South London) at King's College Hospital NHS Foundation. Ann McNeill and Deborah Robson were also funded by the Environmental Exposures and Health Protection Research Unit led by Imperial College London during the current study.

David Hammond has served as a paid expert witness on behalf of public health authorities in response to legal challenges from tobacco, vaping and cannabis companies.

Maciej L Goniewicz serves as a member of the scientific advisory board to Johnson & Johnson.

Kirstie Soar has acted as a paid consultant for the National Centre for Smoking Cessation and Training, ThriveTribe, who deliver stop smoking services, and Pharmastrat Ltd, a healthcare consulting company who deliver stop smoking services.

The other authors declare no potential conflicts of interest.

Department of Health and Social Care disclaimer

This publication presents independent research commissioned by the National Institute for Health and Care Research (NIHR). The views and opinions expressed by authors in this publication are those of the authors and do not necessarily reflect those of the NHS, the NIHR, MRC, NIHR Coordinating Centre, the Public Health Research programme or the Department of Health and Social Care.

This synopsis was published based on current knowledge at the time and date of publication. NIHR is committed to being inclusive and will continually monitor best practice and guidance in relation to terminology and language to ensure that we remain relevant to our stakeholders.

Funding

This synopsis presents independent research funded by the National Institute for Health and Care Research (NIHR) Public Health Research programme as award number NIHR130292.

Award publications

This synopsis provided an overview of the research award *What is the impact of vaping on smoking, nicotine intake and toxin exposure among youth in England compared with youth in North America?*

Other articles published as part of this thread are:

Hammond D, Reid JL, Burkhalter R, Bansal Travers M, Gravely S, Hyland A, *et al*. E-cigarette flavors, devices, and brands used by youths before and after partial flavor restrictions in the United States: Canada, England, and the United States, 2017–2020. *Am J Public Health* 2022;**112**:1014–24. https://doi.org/10.2105/AJPH.2022.306780

East K, Taylor EV, Simonavičius E, Reid JL, Burkhalter R, McNeill A, Hammond D. Use of 'Elf Bar' among youth and young adults who currently vape in England: cross-sectional associations with demographics, dependence indicators and reasons for use. *Addiction* 2025;**120**:414–22. https://doi.org/10.1111/add.16463

Hammond D, Reid JL, Goniewicz ML, McNeill A, O'Connor RJ, Corsetti D, *et al*. Nicotine exposure from smoking tobacco and vaping among adolescents in Canada, England and the US. *JAMA Netw Open* 2025;**8**:e2462544. https://doi.org/0.1001/jamanetworkopen.2024.62544

Hammond D, Reid JL, Goniewicz ML, McNeill A, O'Connor RJ, Corsetti D, et al. Biomarkers of toxicant exposure among youth in Canada, England and the United States who vape and/or smoke tobacco or do neither. *Cancer Epidemiol Biomarkers Prev* 2025;**34**:815–24. https://doi.org/10.1158/1055-9965.EPI-24-1338

Taylor EV, Robson D, Brose LS, Hammond D, Reid JL, Goniewicz ML, *et al*. Nicotine and tobacco specific nitrosamine exposure among youth in England who smoke cigarettes and/or vape. Tob *Control* 2025 Dec 24:tc-2024-059230. https://doi.org/10.1136/tc-2024-059230. Epub ahead of print. Erratum in: *Tob Control* 2025 Dec 25:tc-2024-059230corr1. https://doi.org/10.1136/tc-2024-059230corr1

Brose LS, Reid JL, Robson D, McNeill A, Hammond D. Associations between vaping and self-reported respiratory symptoms in young people in Canada, England and the US. *BMC Med* 2024;**22**:213. https://doi.org/10.1186/s12916-024-03428-6

For more information about this research please view the award page (www.fundingawards.nihr.ac.uk/award/NIHR130292).

Additional outputs

Conference presentations

Hammond D, Reid JL, Goniewicz M, McNeill A, O'Connor RJ, Corsetti D, *et al*. *Biomarkers of Exposure among Youth Smokers, Vapers, and Dual Users: Associations with Product Characteristics and Differences across Countries*. Society for Research on Nicotine & Tobacco (SRNT) Annual Conference, San Antonio, TX, 3 March 2023.

Brose L, Reid JL, McNeill A, Hammond D. *Associations between Nicotine Vaping and Respiratory Symptoms in Young People in Canada, England and the US*. Society for Research on Nicotine and Tobacco – Europe (SRNT-E) meeting, 12 September 2023, London, UK.

Taylor E, Reid JL, Robson D, Goniewicz M, Hammond D, McNeill A. *Nicotine Exposure among Youth in England Who Vaped and/or Smoked, Including by Vaping Product Characteristics*. SRNT, Annual Conference, New Orleans, LA, 2025.

About this synopsis

The contractual start date for this research was in July 2020. This article began editorial review in February 2025 and was accepted for publication in August 2025. The authors have been wholly responsible for all data collection, analysis and interpretation, and for writing up their work. The Public Health Research editors and publisher have tried to ensure the accuracy of the authors' article and would like to thank the reviewers for their constructive comments on the draft document. However, they do not accept liability for damages or losses arising from material published in this article.

Copyright

Copyright © 2026 McNeill *et al*. This work was produced by McNeill *et al*. under the terms of a commissioning contract issued by the Secretary of State for Health and Social Care. This is an Open Access publication distributed under the terms of the Creative Commons Attribution CC BY 4.0 licence, which permits unrestricted use, distribution, reproduction and adaptation in any medium and for any purpose provided that it is properly attributed. See: https://creativecommons.org/licenses/by/4.0/. For attribution the title, original author(s), the publication source – NIHR Journals Library, and the DOI of the publication must be cited.

Published by the NIHR Journals Library (www.journalslibrary.nihr.ac.uk), produced by Newgen Digitalworks Pvt Ltd, Chennai, India (www.newgen.co).

Copyright and credit statement

Every effort has been made to obtain the necessary permissions for reproduction, to credit original sources appropriately and to respect copyright requirements. However, despite our diligence, we acknowledge the possibility of unintentional omissions or errors and we welcome notifications of any concerns regarding copyright or permissions.

List of abbreviations

24MPhMA	2,4-Dimethylphenyl mercapturic acid, xylene
2CaHEMA	2-Carbamoyl-2-hydroxyethyl mercapturic acid, acrylamide
2CyEMA	2-Cyanoethyl mercapturic acid, acrylonitrile
3-HC	trans-3'-Hydroxycotinine, nicotine
3-HPMA	3-hydroxypropyl mercapturic acid, acrolein
BzMA	benzyl mercapturic acid, toluene
COVID	coronavirus disease
COVID-19	coronavirus disease discovered in 2019
DHSC	Department of Health and Social Care
ID	identification
ITC	International Tobacco Control

NIH	National Institutes of Health
NIHR	National Institute for Health and Care Research
NNAL	4-(methylnitrosamino)-1-(3-pyridyl)-1-butanol
PAH	polycyclic aromatic hydrocarbon
PhMA	phenyl mercapturic acid
SDD	Smoking, Drinking and Drug Use survey
SGDP	Social and Genetic and Developmental Psychiatry
SRNT	Society for Research on Nicotine and Tobacco
TNE	total nicotine equivalent
TSNA	tobacco-specific nitrosamine
UKHSA	UK Health Security Agency
VOC	volatile organic compound

References

1. Hammond D, Reid JL, Rynard VL, Fong GT, Cummings KM, McNeill A, *et al*. Prevalence of vaping and smoking among adolescents in Canada, England, and the United States: repeat national cross sectional surveys. *BMJ* 2019;**365**:l2219. https://doi.org/10.1136/bmj.l2219
2. Cole AG, Aleyan S, Battista K, Leatherdale ST. Trends in youth e-cigarette and cigarette use between 2013 and 2019: insights from repeat cross-sectional data from the COMPASS study. *Can J Public Health* 2021;**112**:60–9. https://doi.org/10.17269/s41997-020-00389-0
3. McNeill A, Brose LS, Calder R, Bauld L, Robson D. *Vaping in England: An Evidence Update February 2019*. Public Health England; 2019. URL: https://assets.publishing.service.gov.uk/media/5d3ad8baed915d0d05cb187d/Vaping_in_England_an_evidence_update_February_2019.pdf (accessed 14 August 2025).
4. U.S. Food and Drug Administration. *FDA Regulation of Electronic Nicotine Delivery Systems and Investigation of Vaping Illnesses. Testimony before the House Committee on Energy and Commerce, Subcommittee on Oversight and Investigations*. 2019. URL: www.fda.gov/news-events/congressional-testimony/fda-regulation-electronic-nicotine-delivery-systems-and-investigation-vaping-illnesses-09252019 (accessed 14 August 2025).
5. Health Canada. *Vaping Products Backgrounder*. 2018. URL: www.canada.ca/en/health-canada/services/smoking-tobacco/regulating-tobacco-vaping.html (accessed 29 November 2025).
6. Hammond D, White CM, Czoli CD, Martin CL, Magennis P, Shiplo S. Retail availability and marketing of electronic cigarettes in Canada. *Can J Public Health* 2015;**106**:e408–12. https://doi.org/10.17269/cjph.106.5105
7. Talih S, Salman R, El-Hage R, Karam E, Karaoghlanian N, El-Hellani A, *et al*. Characteristics and toxicant emissions of JUUL electronic cigarettes. *Tob Control* 2019;**28**:678–80. https://doi.org/10.1136/tobaccocontrol-2018-054616
8. Phillips-Waller A, Przulj D, Smith KM, Pesola F, Hajek P. Nicotine delivery and user reactions to Juul EU (20 mg/ml) compared with Juul US (59 mg/ml), cigarettes and other e-cigarette products. *Psychopharmacology* 2021;**238**:825–31. https://doi.org/10.1007/s00213-020-05734-2
9. Hammond D. *ITC Youth Tobacco and Vaping Survey*. University of Waterloo; 2025. URL: http://davidhammond.ca/projects/e-cigarettes/itc-youth-tobacco-ecig/ (accessed 13 August 2025).
10. McNeill A, Robson D, Brose LS. *What Is the Impact of Vaping on Smoking, Nicotine Intake and Toxin Exposure among Youth in England Compared with Youth in North America?* NIHR; 2023. URL: https://fundingawards.nihr.ac.uk/award/NIHR130292 (accessed 13 August 2025).
11. STROBE. *Checklist*. 2025. URL: www.strobe-statement.org/checklists/ (accessed 13 August 2025).
12. Jarvis MJ. Why people smoke. *BMJ* 2004;**328**:277–9. https://doi.org/10.1136/bmj.328.7434.277
13. Gika HG, Theodoridis GA, Earll M, Snyder RW, Sumner SJ, Wilson ID. Does the mass spectrometer define the marker? A comparison of global metabolite profiling data generated simultaneously via UPLC-MS on two different mass spectrometers. *Anal Chem* 2010;**82**:8226–34. https://doi.org/10.1021/ac1016612
14. Tevis DS, Flores SR, Kenwood BM, Bhandari D, Jacob P 3rd, Liu J, *et al*. Harmonization of acronyms for volatile organic compound metabolites using a standardized naming system. *Int J Hyg Environ Health* 2021;**235**:113749. https://doi.org/10.1016/j.ijheh.2021.113749
15. Tevis DS, Flores SR, Kenwood BM, Bhandari D, Jacob P 3rd, Liu J, *et al*. Harmonization of acronyms for volatile organic compound metabolites using a standardized naming system. *Int J Hyg Environ Health*

2023;**247**:114030. [Corrigendum published in *Int J Hygiene Environ Health* 2021;**235**:113749.] https://doi.org/10.1016/j.ijheh.2022.114030

16. Hammond D, Reid JL, Goniewicz ML, McNeill A, O'Connor RJ, Corsetti D, *et al*. Nicotine exposure from smoking tobacco and vaping among adolescents. *JAMA Netw Open* 2025;**8**:e2462544. https://doi.org/10.1001/jamanetworkopen.2024.62544
17. Taylor E, Robson D, Brose LS, Hammond D, Reid JL, Goniewicz ML, *et al*. Nicotine and Tobacco Specific Nitrosamine Exposure among Youth in England Who Smoke Cigarettes and/or Vape. *Tob Control* 2025 Dec 24:tc-2024-059230. https://doi.org/10.1136/tc-2024-059230. Epub ahead of print. Erratum in: *Tob Control* 2025 Dec 25:tc-2024-059230corr1. https://doi.org/10.1136/tc-2024-059230corr1
18. Benowitz NL. Nicotine addiction. *N Engl J Med* 2010;**362**:2295–303. https://doi.org/10.1056/NEJMra0809890
19. U.S. Food and Drug Administration. *Harmful and Potentially Harmful Constituents in Tobacco Products and Tobacco Smoke: Established List*. 2012. URL: www.fda.gov/tobacco-products/rules-regulations-and-guidance-related-tobacco-products/harmful-and-potentially-harmful-constituents-tobacco-products-and-tobacco-smoke-established-list (accessed 14 August 2025).
20. Benowitz NL, Burbank AD. Cardiovascular toxicity of nicotine: implications for electronic cigarette use. *Trends Cardiovasc Med* 2016;**26**:515–23. https://doi.org/10.1016/j.tcm.2016.03.001
21. Hammond D, Reid JL, Goniewicz ML, McNeill A, O'Connor RJ, Corsetti D, *et al*. Biomarkers of toxicant exposure among youth in Canada, England and the United States who vape and/or smoke tobacco or do neither. *Cancer Epidemiol Biomarkers Prev* 2025;**34**:815–24. https://doi.org/10.1158/1055-9965.EPI-24-1338
22. International Agency for Research on Cancer. *Agents Classified by the IARC Monographs, Volumes 1–136*. URL: https://monographs.iarc.who.int/agents-classified-by-the-iarc/ (accessed 31 January 2025).
23. U.S. Environmental Protection Agency. *Inhalation Health Effect Reference Values for Xylene: All Isomers*. 2012. URL: https://oaspub.epa.gov/eims/eimscomm.getfile%3Fp_download_id%3D512651 (accessed 14 August 2025).
24. Statista. *Reach of Leading Social Networking Sites Used by Those Aged 15 to 25 in the United Kingdom (UK) as of 3rd Quarter 2020*. 2025. URL: www.statista.com/statistics/1059462/social-media-usage-uk-age/ (accessed 13 August 2025).
25. Hype T. *The Hype King's College London*. URL: https://hypekcl.com/ (accessed 13 August 2025).
26. NHS Health Research Authority. *Research Involving Children*. 2024. URL: www.hra.nhs.uk/planning-and-improving-research/policies-standards-legislation/research-involving-children/ (accessed 14 August 2025).
27. Hammond D, Reid JL, Burkhalter R, Bansal Travers M, Gravely S, Hyland A, *et al*. e-Cigarette flavors, devices, and brands used by youths before and after partial flavor restrictions in the United States: Canada, England, and the United States, 2017–2020. *Am J Public Health* 2022;**112**:1014–24. https://doi.org/10.2105/AJPH.2022.306780
28. East K, Taylor EV, Simonavicius E, Reid JL, Burkhalter R, McNeill A, Hammond D. Use of 'Elf Bar' among youth and young adults who currently vape in England: cross-sectional associations with demographics, dependence indicators and reasons for use. *Addiction* 2025;**120**:414–22. https://doi.org/10.1111/add.16463
29. Hammond D, Reid J, McNeill A, Goniewicz ML, O'Connor R, Corsetti D, *et al*. *Biomarkers of Exposure among Youth Who Smoke, Vape, and 'Dual Use': Associations with Product Characteristics and Potential Differences across Countries*. OSF Registries; 2023. URL: https://osf.io/9z3cu (accessed 13 August 2025).
30. Taylor E, Brose LS, McNeill A, Robson D, Hammond D, Reid J. *Biomarkers of Exposure among Youth Who Smoke, Vape, Dual Use or Neither Vape nor Smoke in England*. OSF Registries; 2024. URL: https://osf.io/xuw3n (accessed 13 August 2025).
31. Brose LS, Reid JL, Robson D, McNeill A, Hammond D. Associations between vaping and self-reported respiratory symptoms in young people in Canada, England and the US. *BMC Med* 2024;**22**:213. https://doi.org/10.1186/s12916-024-03428-6
32. Brose LS, McNeill A, Robson D, Reid JL, Hammond D. *Vaping and Respiratory Symptoms ITC Youth*. OSF registries; 2022. URL: https://osf.io/9z3cu (accessed 13 August 2025).
33. Action on Smoking and Health (ASH). *Use of Vapes (e-Cigarettes) among Young People in Great Britain*. 2024. URL: https://ash.org.uk/uploads/Use-of-vapes-among-young-people-in-Great-Britain-2024.pdf (accessed 15 August 2025).
34. Action on Smoking and Health (ASH). *Use of Vapes (e-Cigarettes) among Adults in Great Britain*. 2024. URL: https://ash.org.uk/uploads/Use-of-vapes-among-adults-in-Great-Britain-2024.pdf (accessed 15 August 2025).

35. Gravely S, Cummings KM, Hammond D, Lindblom E, Smith DM, Martin N, *et al*. The association of e-cigarette flavors with satisfaction, enjoyment, and trying to quit or stay abstinent from smoking among regular adult vapers from Canada and the United States: findings from the 2018 ITC four country smoking and vaping survey. *Nicotine Tob Res* 2020;**22**:1831–41. https://doi.org/10.1093/ntr/ntaa095

36. Leventhal AM, Goldenson NI, Cho J, Kirkpatrick MG, McConnell RS, Stone MD, *et al*. Flavored e-cigarette use and progression of vaping in adolescents. *Pediatrics* 2019;**144**:e20190789. https://doi.org/10.1542/peds.2019-0789

37. Lindson N, Livingstone-Banks J, Butler AR, Levy DT, Barnett P, Theodoulou A, *et al*. An update of a systematic review and meta-analyses exploring flavours in intervention studies of e-cigarettes for smoking cessation. *Addiction* 2024;**120**:770–8. https://doi.org/10.1111/add.16736

38. Livingstone-Banks J, Travis N, Conde M, Chen YC, Zi P, Jarman H, *et al*. The impacts of e-cigarette flavours: an overview of systematic reviews. *Addiction* 2025;**120**:1327–44. https://doi.org/10.1111/add.70017

39. Department of Health and Social Care (DHSC). *The Tobacco and Vapes Bill 2024*. 2024. URL: www.gov.uk/government/collections/the-tobacco-and-vapes-bill-2024 (accessed 15 August 2025).

40. Tam J, Jimenez-Mendoza E, Buckell J, Sindelar J, Meza R. Responses to real-world and hypothetical e-cigarette flavor bans among US young adults who use flavored e-cigarettes. *Nicotine Tob Res* 2024;**26**:1113–7. https://doi.org/10.1093/ntr/ntad258

41. Smith MJ, MacKintosh AM, Ford A, Hilton S. Youth's engagement and perceptions of disposable e-cigarettes: a UK focus group study. *BMJ Open* 2023;**13**:e068466. https://doi.org/10.1136/bmjopen-2022-068466

42. McNeill A, Simonavicius E, Brose L, Taylor E, East K, Zuikova E, *et al*. *Nicotine Vaping in England: An Evidence Update Including Health Risks and Perceptions, 2022*. London: Office for Health Improvement and Disparities; 2022. URL: https://assets.publishing.service.gov.uk/media/633469fc8fa8f5066d28e1a2/Nicotine-vaping-in-England-2022-report.pdf (accessed 15 August 2025).

43. Department of Health and Social Care (DHSC). *Minister Neil O'Brien Speech on Achieving a Smokefree 2030: Cutting Smoking and Stopping Kids Vaping*. 2023. URL: www.gov.uk/government/speeches/minister-neil-obrien-speech-on-achieving-smokefree-2030-cutting-smoking-and-stopping-kids-vaping (accessed 15 August 2025).

44. Boykan R, Messina CR, Chateau G, Eliscu A, Tolentino J, Goniewicz ML. Self-reported use of tobacco, e-cigarettes, and Marijuana versus urinary biomarkers. *Pediatrics* 2019;**143**:e20183531. https://doi.org/10.1542/peds.2018-3531

45. Boykan R, Goniewicz ML, Messina CR. Evidence of nicotine dependence in adolescents who use JUUL and similar pod devices. *Int J Environ Res Public Health* 2019;**16**:2135. https://doi.org/10.3390/ijerph16122135

46. Chaffee BW, Jacob P, Couch ET, Benowitz NL. Exposure to a tobacco-specific carcinogen among adolescent smokeless tobacco users in rural California, United States. *Nicotine Tob Res* 2020;**22**:1764–71. https://doi.org/10.1093/ntr/ntz147

47. Vogel EA, Prochaska JJ, Ramo DE, Andres J, Rubinstein ML. Adolescents' e-cigarette use: increases in frequency, dependence, and nicotine exposure over 12 months. *J Adolesc Health* 2019;**64**:770–5. https://doi.org/10.1016/j.jadohealth.2019.02.019

48. Cho YJ, Mehta T, Hinton A, Sloan R, Nshimiyimana J, Tackett AP, *et al*. e-Cigarette nicotine delivery among young adults by nicotine form, concentration, and flavor: a crossover randomized clinical trial. *JAMA Netw Open* 2024;**7**:e2426702. https://doi.org/10.1001/jamanetworkopen.2024.26702

49. Christen SE, Hermann L, Bekka E, Vonwyl C, Hammann F, van der Velpen V, *et al*. Pharmacokinetics and pharmacodynamics of inhaled nicotine salt and free-base using an e-cigarette: a randomized crossover study. *Nicotine Tob Res* 2024;**26**:1313–21. https://doi.org/10.1093/ntr/ntae074

50. Oliveri D, Liang Q, Sarkar M. Real-world evidence of differences in biomarkers of exposure to select harmful and potentially harmful constituents and biomarkers of potential harm between adult e-vapor users and adult cigarette smokers. *Nicotine Tob Res* 2020;**22**:1114–22. https://doi.org/10.1093/ntr/ntz185

51. Rostron BL, Coleman B, Cheng YC, Kimmel HL, Oniyide O, Wang L, Chang CM. Nicotine exposure by device type among adult electronic nicotine delivery system users in the population assessment of tobacco and health study, 2015–2016. *Cancer Epidemiol Biomarkers Prev* 2020;**29**:1968–72. https://doi.org/10.1158/1055-9965.EPI-20-0317

52. Rubinstein ML, Delucchi K, Benowitz NL, Ramo DE. Adolescent exposure to toxic volatile organic chemicals from e-cigarettes. *Pediatrics* 2018;**141**:e20173557. https://doi.org/10.1542/peds.2017-3557

53. Boyd CJ, McCabe SE, Evans-Polce RJ, Veliz PT. Cannabis, vaping, and respiratory symptoms in a probability sample of U.S. youth. *J Adolesc Health*

2021;**69**:149–52. https://doi.org/10.1016/j.jadohealth.2021.01.019

54. Cherian C, Buta E, Simon P, Gueorguieva R, Krishnan-Sarin S. Association of vaping and respiratory health among youth in the population assessment of tobacco and health (PATH) study wave 3. *Int J Environ Res Public Health* 2021;**18**:8208. https://doi.org/10.3390/ijerph18158208

55. Hyland A, Ambrose BK, Conway KP, Borek N, Lambert E, Carusi C, *et al*. Design and methods of the Population Assessment of Tobacco and Health (PATH) study. *Tob Control* 2017;**26**:371–8. https://doi.org/10.1136/tobaccocontrol-2016-052934

56. Stevens ER, Xu S, Niaura R, Cleland CM, Sherman SE, Mai A, *et al*. Youth e-cigarette use and functionally important respiratory symptoms: the Population Assessment of Tobacco and Health (PATH) study waves 3 and 4. *Int J Environ Res Public Health* 2022;**19**:15324. https://doi.org/10.3390/ijerph192215324

57. Tackett AP, Keller-Hamilton B, Smith CE, Hebert ET, Metcalf JP, Queimado L, *et al*. Evaluation of respiratory symptoms among youth e-cigarette users. *JAMA Netw Open* 2020;**3**:e2020671. https://doi.org/10.1001/jamanetworkopen.2020.20671

58. Tattan-Birch H, Jackson SE, Kock L, Dockrell M, Brown J. Rapid growth in disposable e-cigarette vaping among young adults in Great Britain from 2021 to 2022: a repeat cross-sectional survey. *Addiction* 2023;**118**:382–6. https://doi.org/10.1111/add.16044

59. Tackett AP, Urman R, Barrington-Trimis J, Liu F, Hong H, Pentz MA, *et al*. Prospective study of e-cigarette use and respiratory symptoms in adolescents and young adults. *Thorax* 2023;**79**:163–8. https://doi.org/10.1136/thorax-2022-218670

60. Public Health England. *Indoor Air Quality Guidelines for Selected Volatile Organic Compounds (VOCs) in the UK*. 2019. URL: https://assets.publishing.service.gov.uk/media/5d7a2912ed915d522e4164a5/VO__statement_Final_12092019_CS__1_.pdf (accessed 15 August 2025).

61. Fix BV, O'Connor R, Hammond D, King B, McNeill A, Thrasher J, *et al*. ITC 'spit and butts' pilot study: the feasibility of collecting saliva and cigarette butt samples from smokers to evaluate policy. *Nicotine Tob Res* 2010;**12**:185–90. https://doi.org/10.1093/ntr/ntp191

62. Block AC, Schneller LM, Leigh NJ, Heo J, Goniewicz ML, O'Connor RJ. Heavy metals in ENDS: a comparison of open versus closed systems purchased from the USA, England, Canada and Australia. *Tob Control* 2025;**34**:28–33. https://doi.org/10.1136/tc-2023-057932

63. King's College London. *Vaping and Smoking among People Aged 16–19 in England*. King's College London; 2023. URL: www.kcl.ac.uk/research/vaping-smoking-young-people-england (accessed 14 August 2025).

64. NHS Digital. *Smoking, Drinking and Drug Use among Young People in England, 2023*. 2023. URL: https://digital.nhs.uk/data-and-information/publications/statistical/smoking-drinking-and-drug-use-among-young-people-in-england (accessed 14 Aug 2025).

65. Cadham CJ, Liber AC, Sanchez-Romero LM, Issabakhsh M, Warner KE, Meza R, Levy DT. The actual and anticipated effects of restrictions on flavoured electronic nicotine delivery systems: a scoping review. *BMC Public Health* 2022;**22**:2128. https://doi.org/10.1186/s12889-022-14440-x

66. Freitas-Lemos R, Stein JS, Tegge AN, Kaplan BA, Heckman BW, McNeill A, *et al*. Illegal experimental tobacco marketplace II: effects of vaping product bans – findings from the 2020 International Tobacco Control Project. *Tob Control* 2022;**31**:s214–22. https://doi.org/10.1136/tc-2022-057515

67. Yang Y, Lindblom EN, Ward KD, Salloum RG. Reactions to hypothetical flavor bans among current users of flavored e-cigarettes. *Transl Behav Med* 2023;**13**:533–8. https://doi.org/10.1093/tbm/ibac109

68. Taylor E, Tattan-Birch H, East K. England's disposable vape ban: an inadequate solution to youth vaping with potential unintended consequences. *Addiction* 2025;**120**:402–4. https://doi.org/10.1111/add.16756